EART 22201 Igneous minerals and processes

Magmatism at destructive plate margins



Lesson 8

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Magmatism at destructive plate margins

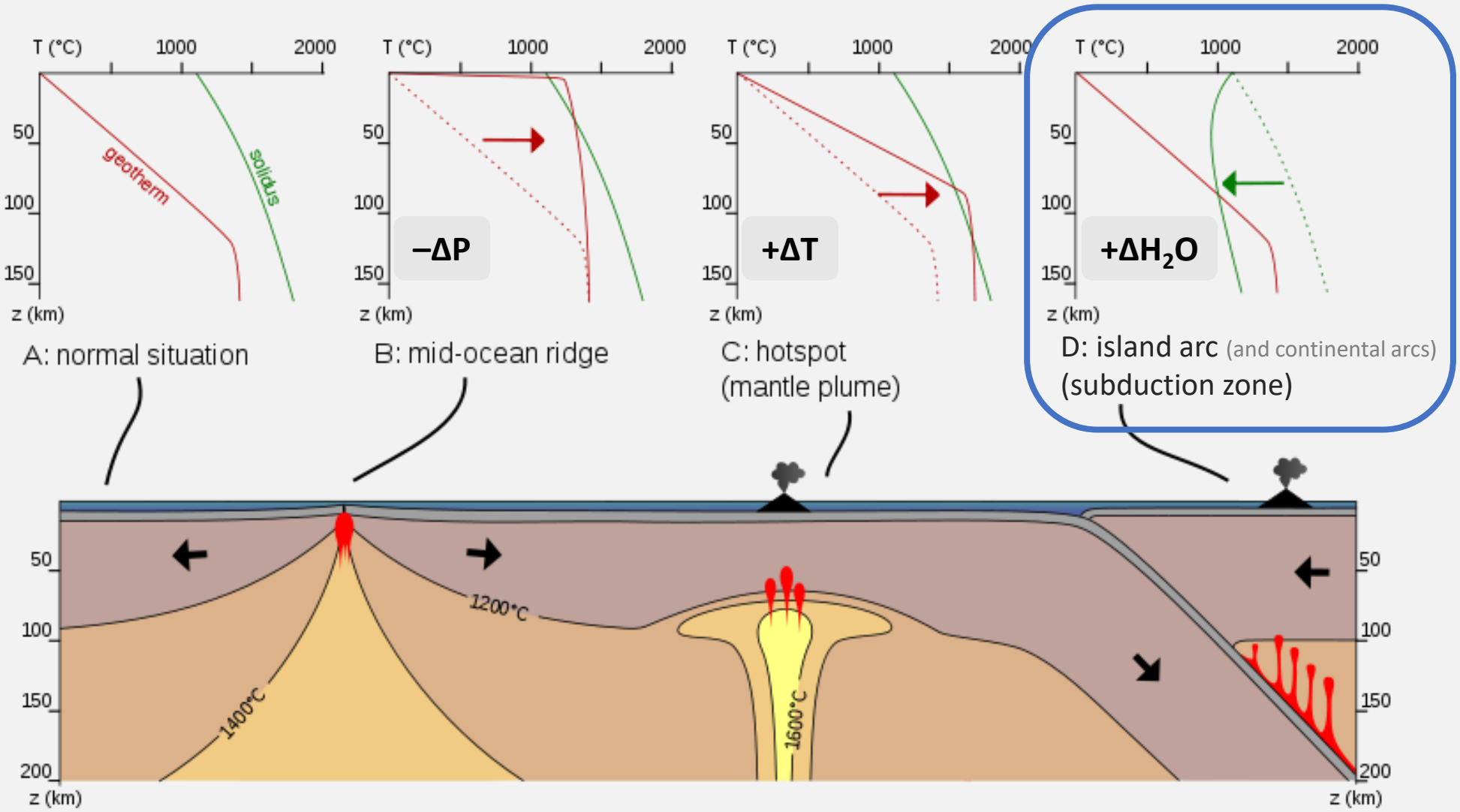
1.1 Subduction zone magmatism

1.2 Flux melting



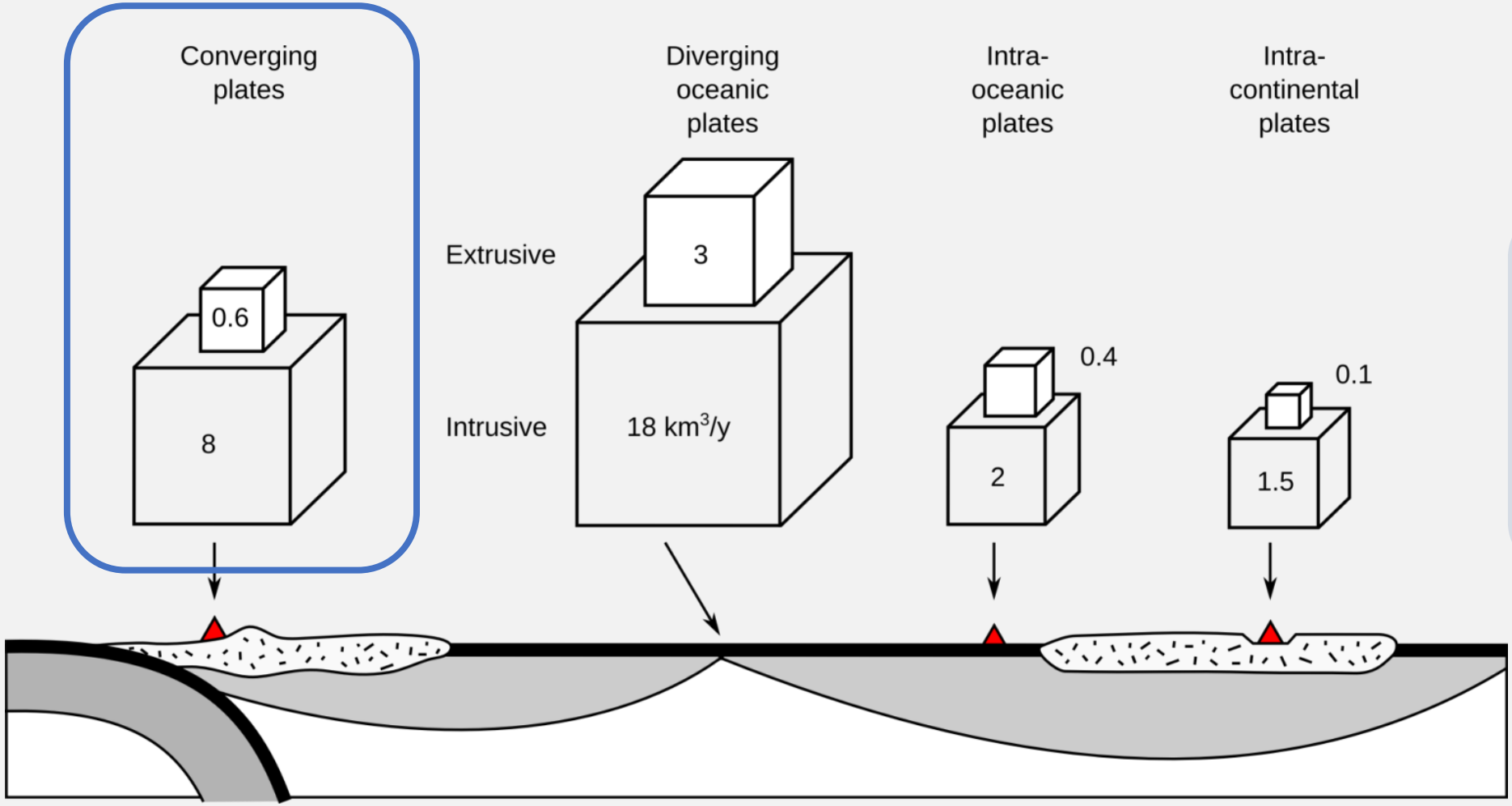
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Where and how are different primary melt compositions produced?



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Global inventory of magma production



Magmatism at destructive plate margins (island arcs plus continental arcs):

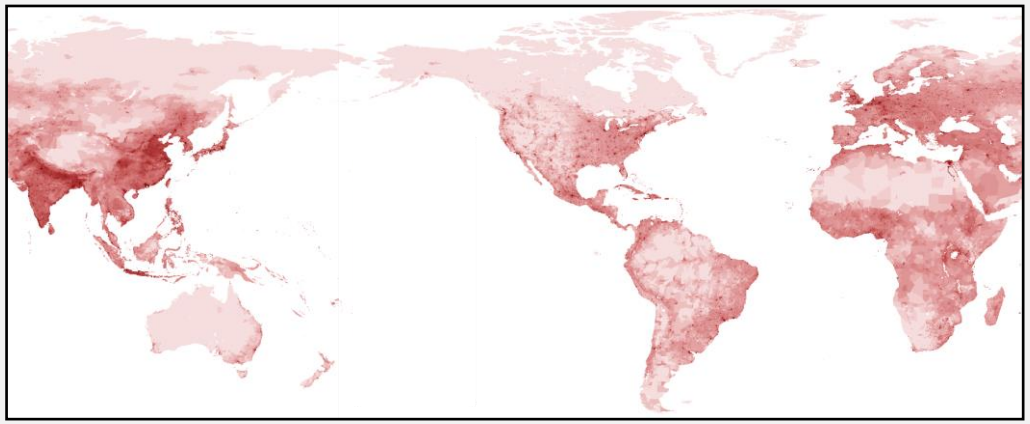
- ~15% of global extrusive magma volume
- ~25% of global intrusive magma volume

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Earth's active subaerial volcanoes are mainly located along destructive plate margins



- Many DPM volcanoes are near to large population centres or densely populated regions
- Potential for large, hazardous, destructive eruptions



Map of global population density

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Magmatism at destructive plate margins

1.1 Subduction zone magmatism

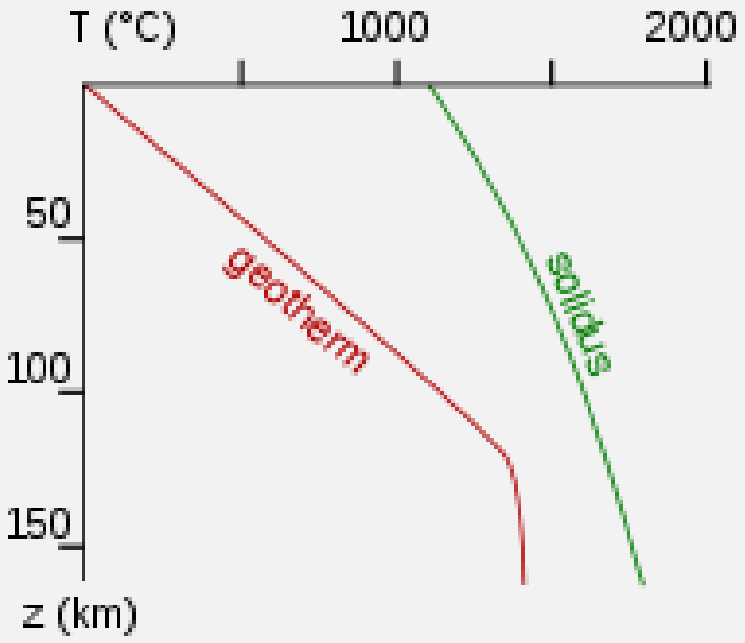
1.2 Flux melting



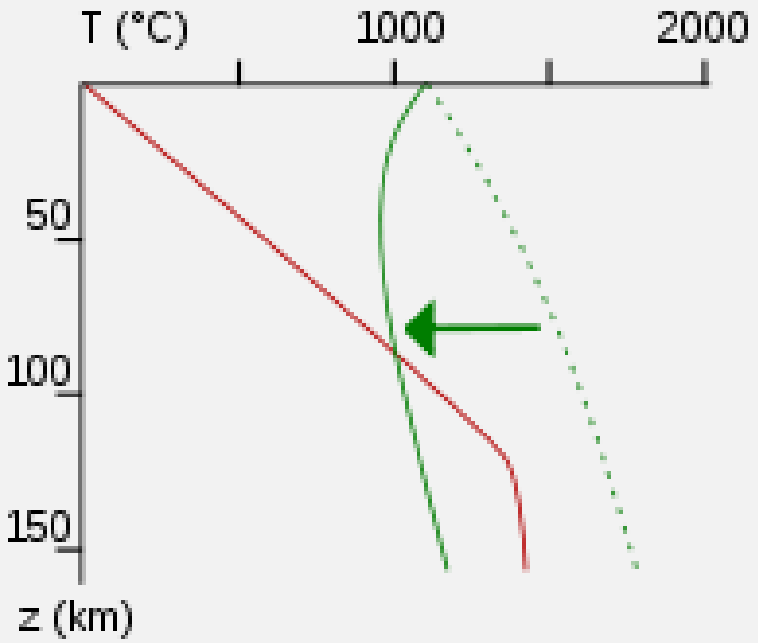
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Flux melting at subduction zones

Dry peridotite solidus



Wet peridotite solidus



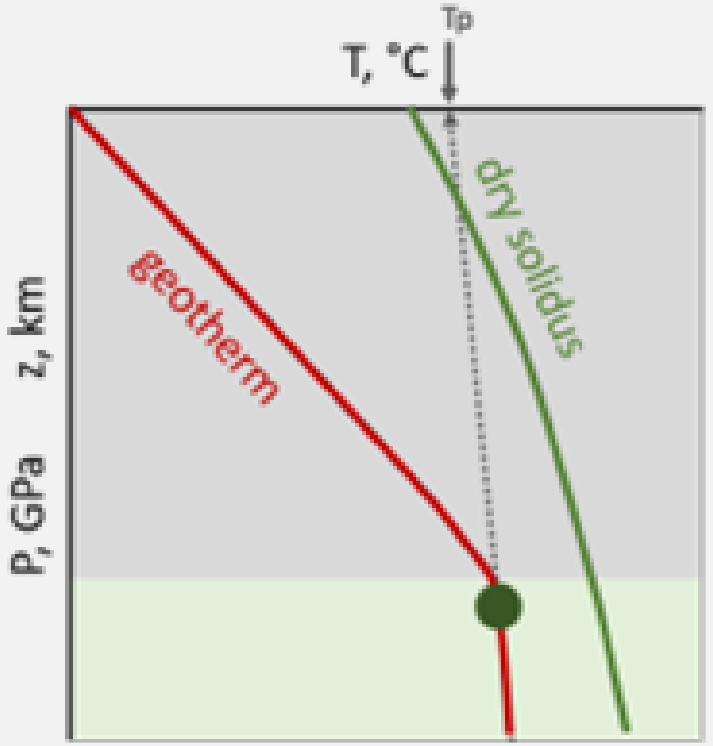
Flux melting (or hydrous melting)

The solidus of mantle peridotite is lowered by the addition of water

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Flux melting at subduction zones



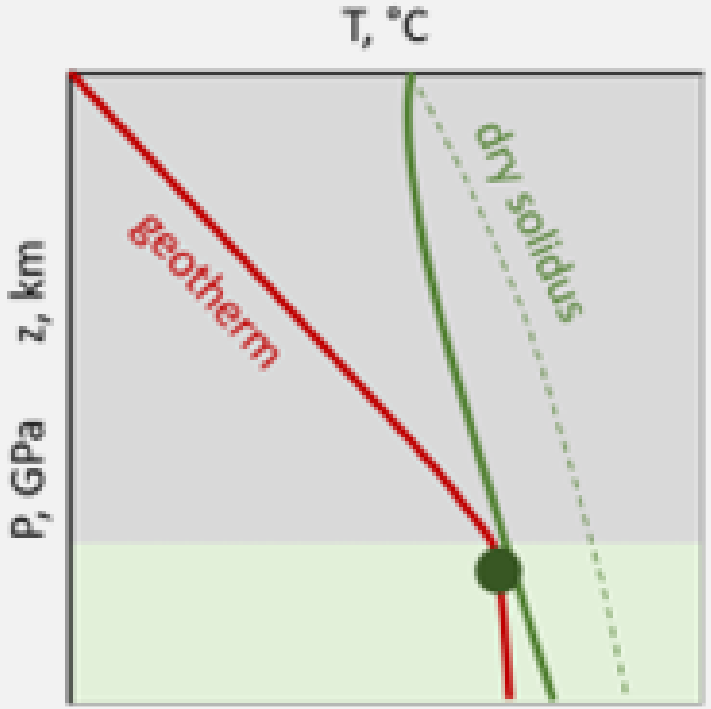


The peridotite solidus is lowered by the addition of water. Eventually the solidus crosses the geotherm, allowing peridotite to melt.

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Flux melting at subduction zones

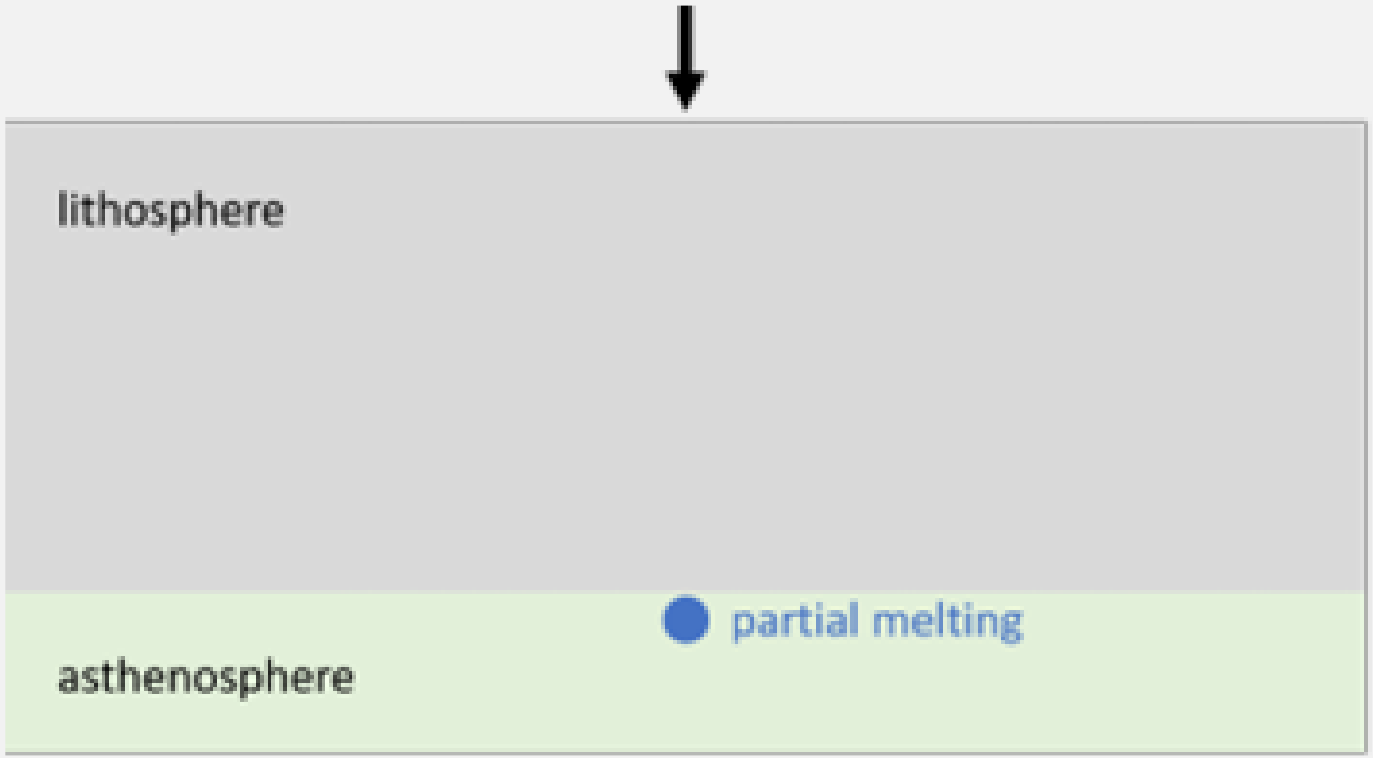


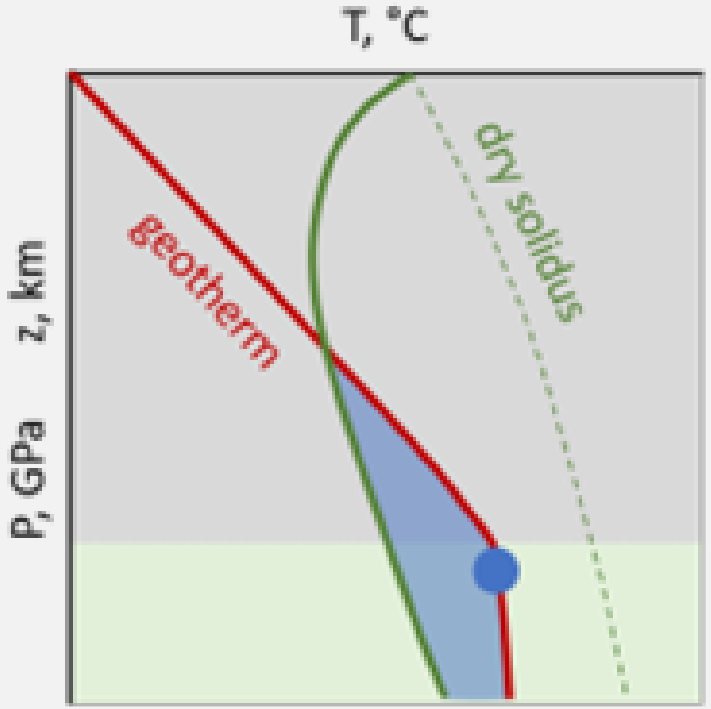


The peridotite solidus is lowered by the addition of water. Eventually the solidus crosses the geotherm, allowing peridotite to melt.

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Flux melting at subduction zones





The peridotite solidus is lowered by the addition of water. Eventually the solidus crosses the geotherm, allowing peridotite to melt.

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Magmatism at destructive plate margins

You should now be able to:

- Explain the concept of flux melting (hydrous melting)
- Construct a P-T diagram to show how the geotherm and solidus intersect to cause flux melting at subduction zones



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Magmatism at destructive plate margins

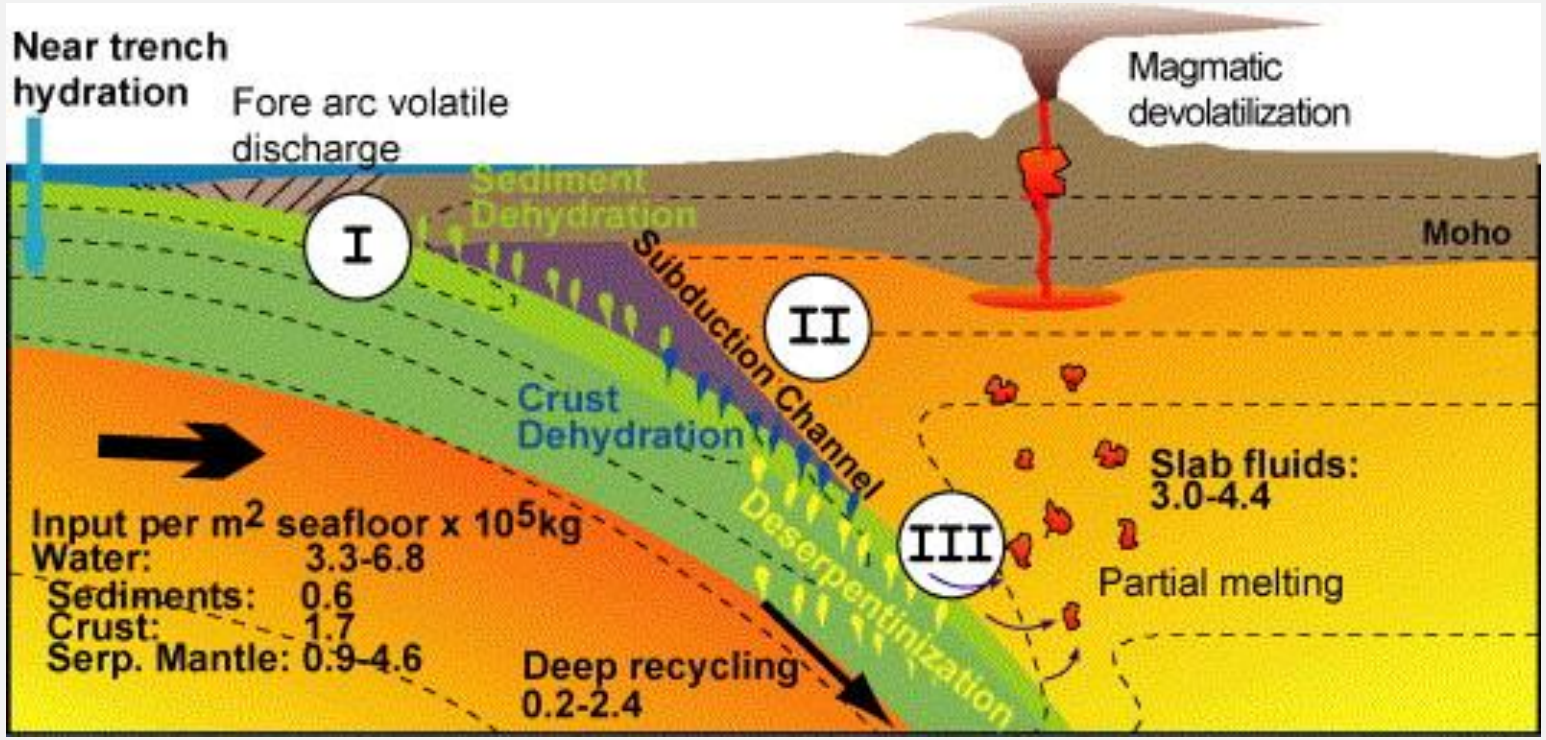
2.1 Where does the water for flux melting come from?

2.2 What melts?



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The water required for flux melting comes from the downgoing slab



(I) Depth <20 km: Release of sedimentary pore fluids. Low temperature so no melting.

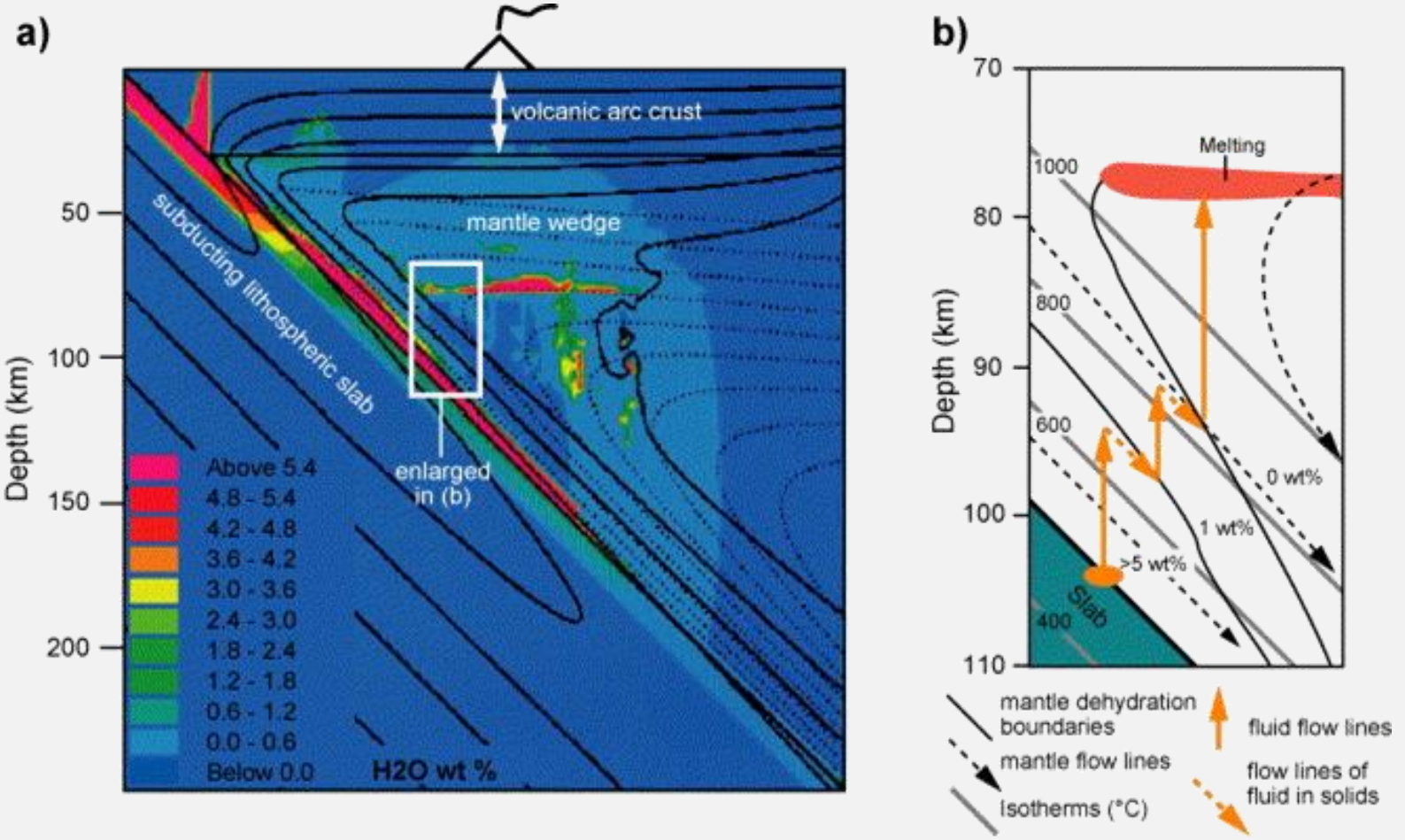
(II) Depth ~20-100 km: Further dehydration of sediments and oceanic crust. Temperature still too low for melting.

(III) Depth >100 km: Hydrous minerals in the subducting oceanic lithosphere break down and release their structurally bound water.

Rüpke et al. (2004) Earth Planet Sci Lett 223:17-34 https://doi.org/10.1016/j.epsl.2004.04.018

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The water required for flux melting comes from the downgoing slab



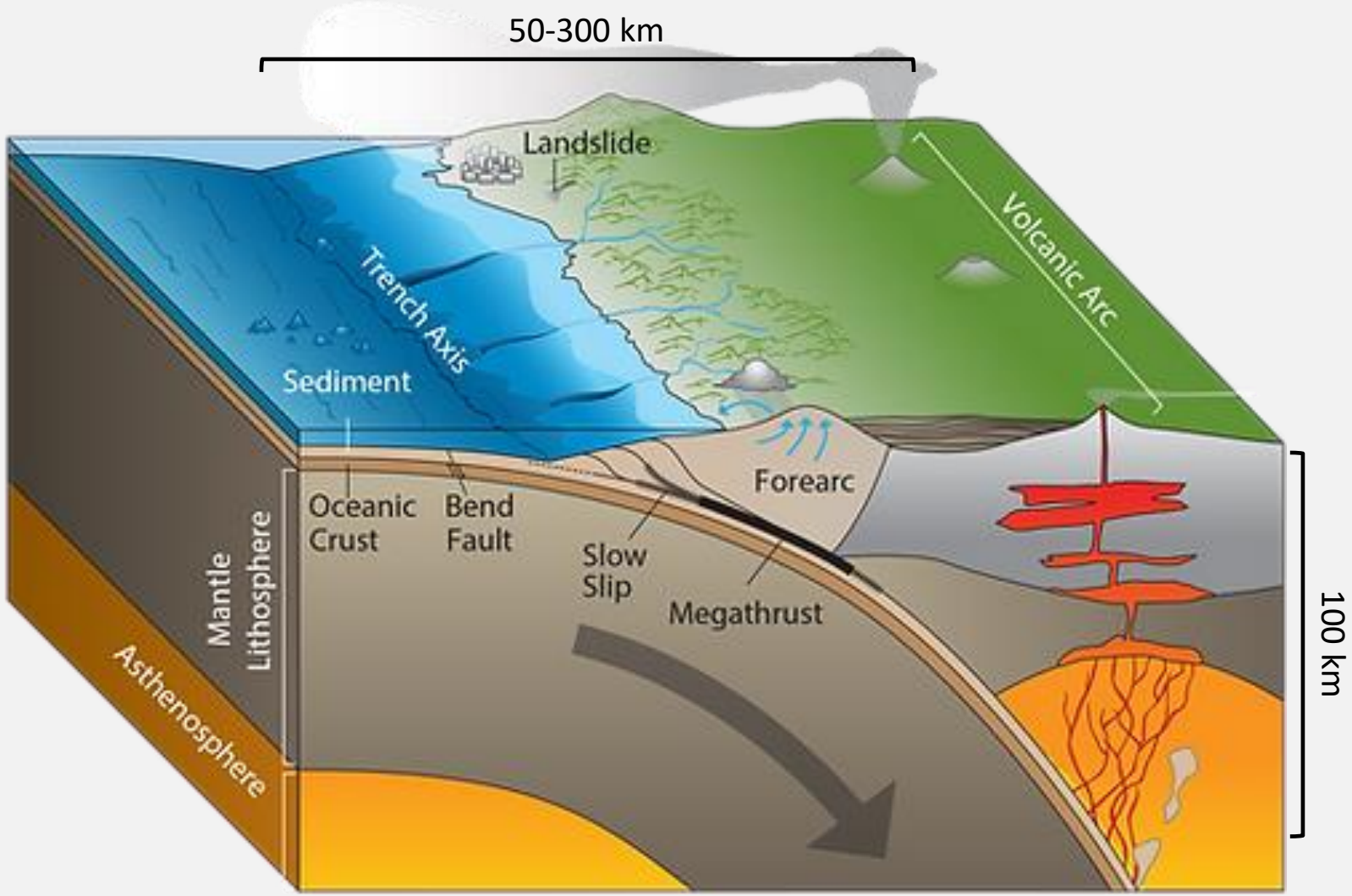
(a) Diagram showing expected water contents within a subduction zone. Colours represent rock H_2O content. In the slab, H_2O decreases with depth due to dehydration, then breakdown of hydrous minerals. The high-H_2O region in the mantle wedge, beneath the volcanic arc, shows where flux melting of mantle peridotite begins.

(b) Diagram showing the schematic path of fluids derived by breakdown of hydrous minerals in the slab. The fluid migrates upwards into the mantle wedge (solid orange arrows), where it reacts with peridotite and forms new hydrous minerals. The hydrated mantle flows downwards (dashed arrows) and then dehydrates, re-releasing the fluid. After multiple hydration-dehydration steps, the fluid reaches a region of the mantle where the temperature is high enough for flux melting to occur.

Manning (2004) Earth Planet Sci Lett 223:1-16 https://doi.org/10.1016/j.epsl.2004.04.030

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Volcanic arc above the locus of slab dehydration and mantle melting

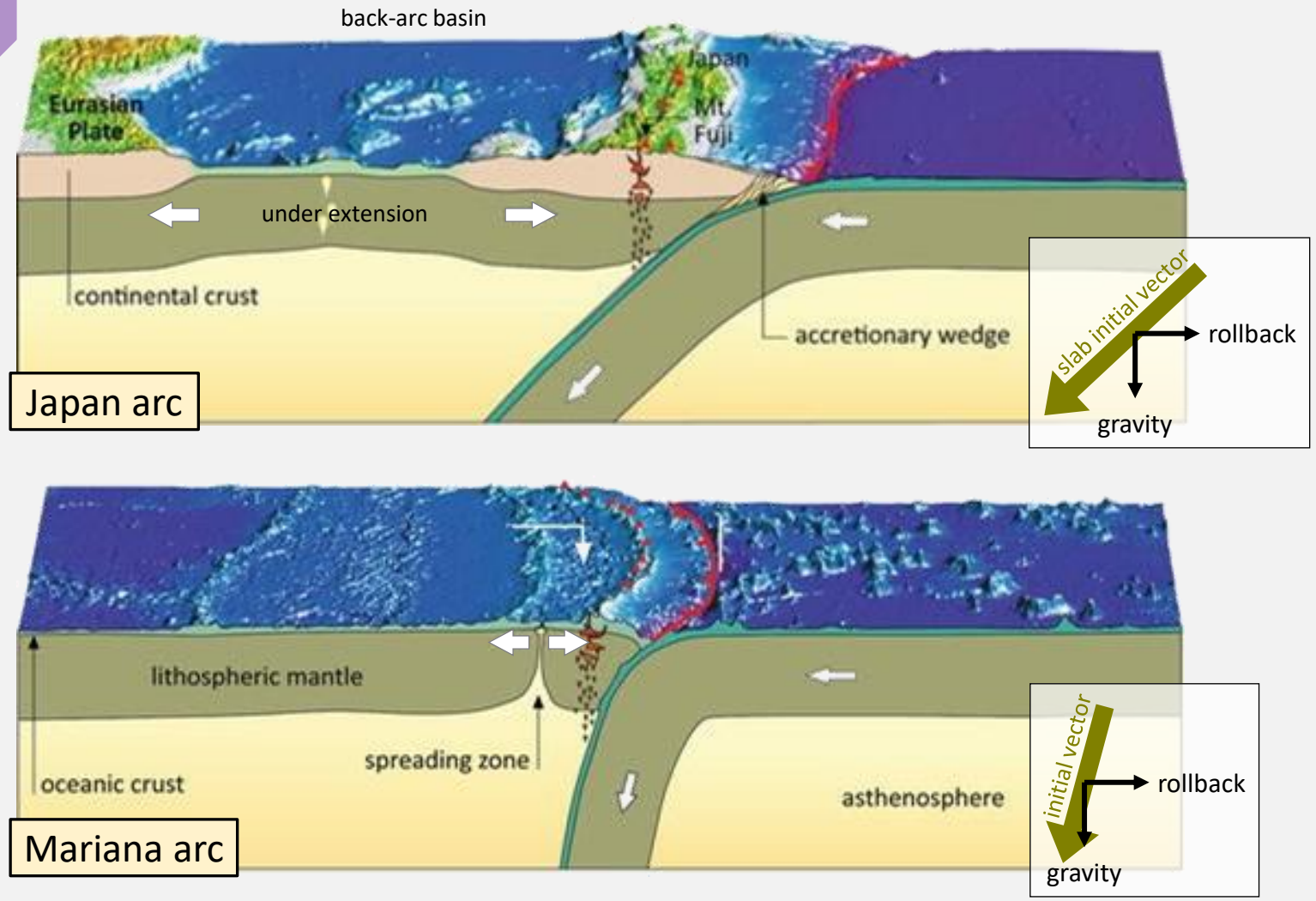


The volcanic arc is located where the depth to the top of the subducting slab is ~100-110 km. This coincides with the breakdown of hydrous minerals in the slab.

The distance between trench and the volcanic arc depends on the slab dip.

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Slab rollback can lead to back-arc spreading and back-arc volcanism



- Cold, dense subducting slab sinks into the asthenospheric mantle. The overriding plate is under extension.
- Extension causes lithospheric thinning behind the volcanic arc, creating a back-arc basin.
- Back-arc volcanism occurs behind the volcanic arc, driven by decompression melting of mantle peridotite.

Analogue model of subduction: video

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Magmatism at destructive plate margins

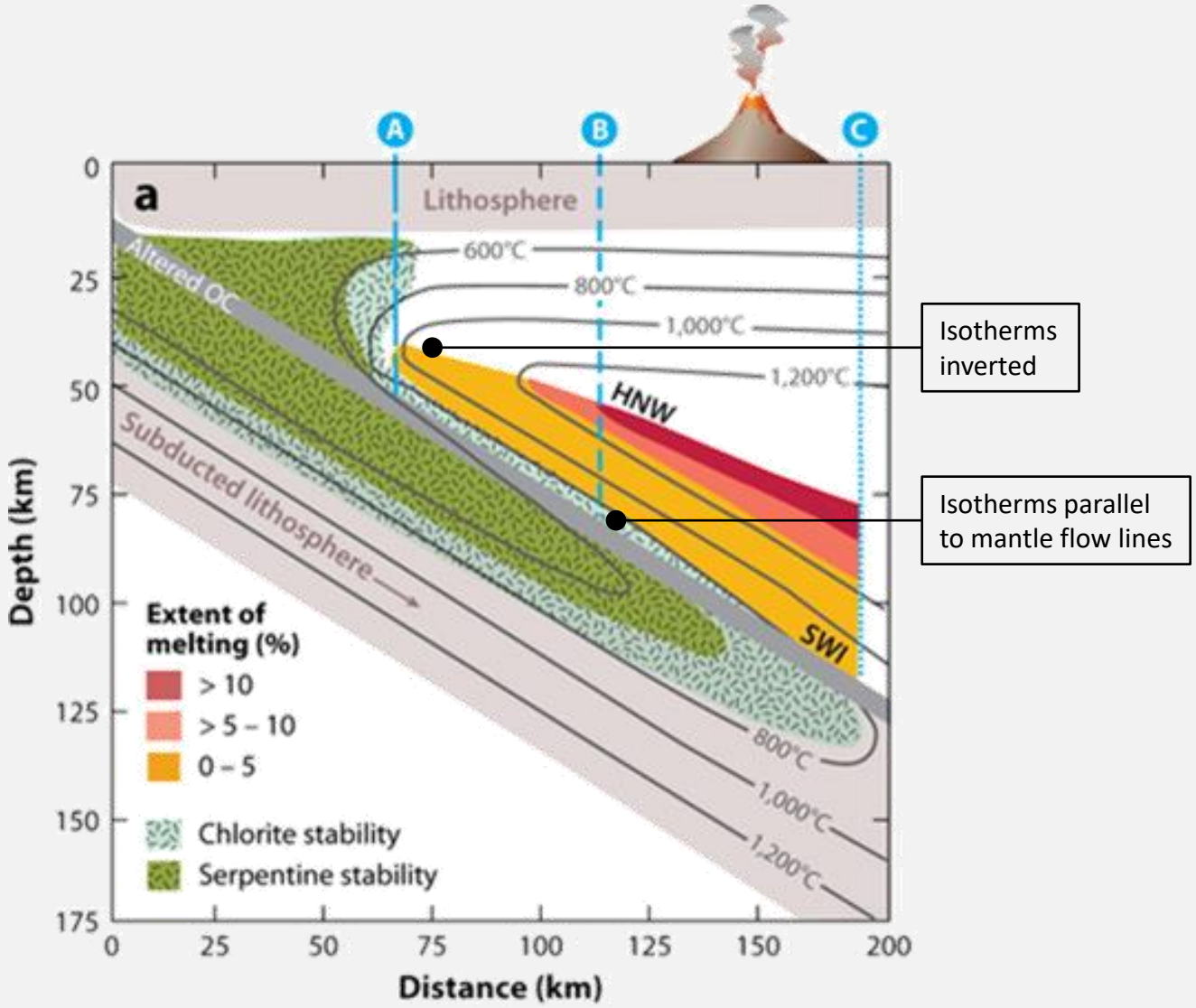
2.1 Where does the water for flux melting come from?

2.2 What melts?



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Flux melting of peridotite occurs in the mantle wedge



Isotherms inverted

Isotherms parallel to mantle flow lines

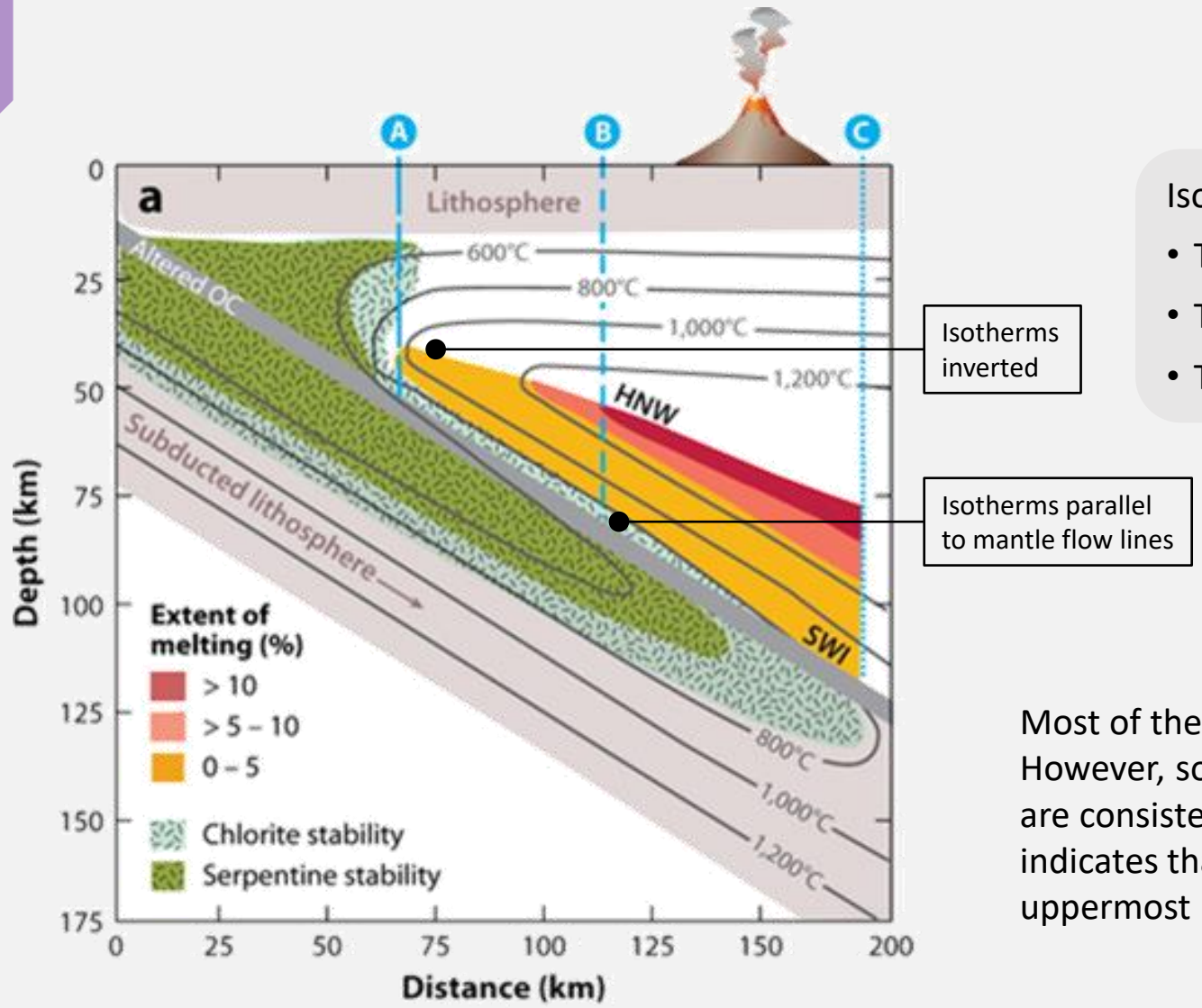
The vast majority of the magma volume at subduction zones is produced by partial melting of asthenospheric sub-arc mantle peridotite in the mantle wedge.

Melting can occur in the garnet and spinel peridotite fields.

The slab is too cold to undergo partial melting.

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Does the slab itself ever melt?



Isotherms inverted

Isotherms parallel to mantle flow lines

Isotherms will be higher (and the system will be hotter) if:

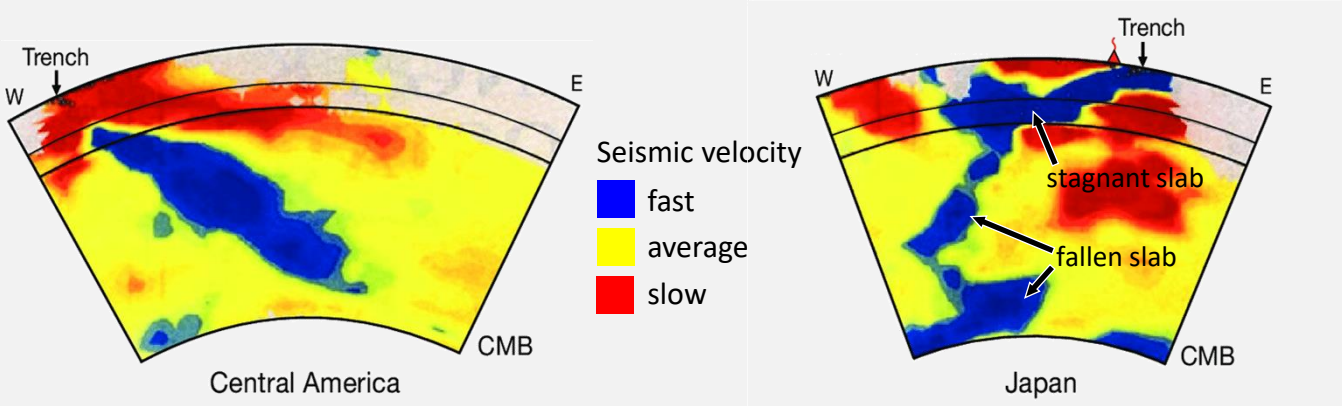
- The convergence rate is slower
- The slab is young and near the ridge, therefore warm
- The arc is young (<50-100 Ma)

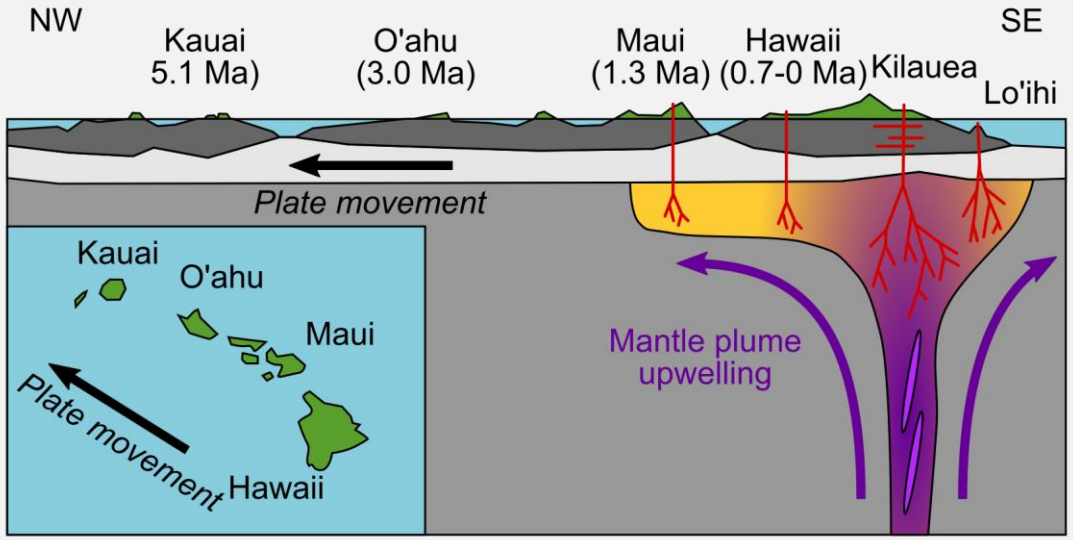
Most of the downgoing slab **remains too cold** to undergo partial melting. However, some subduction zone magmas have chemical signatures that are consistent with partial melting of sediments or oceanic crust. This indicates that, under the right thermal conditions, it is possible for the uppermost parts of the slab to undergo partial melting.

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Unmelted slab material is ultimately recycled into the deep mantle

See also EART 26201 Principles of Geochemistry, Lecture 2





Subducted slabs imaged by seismic tomography

- Some slabs reach the core-mantle boundary
- Some slabs stagnate at the mantle transition zone (~660 km)

Subduction recycling creates heterogeneity in the mantle. This heterogeneity is sampled when mantle plumes carry melts of different mantle types to the surface

- Example: the **Loa** and **Kea** trends on Hawaii

See Week 7

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Magmatism at destructive plate margins

You should now be able to:

- Explain the source of water that drives flux melting at subduction zones
- Sketch and qualitatively describe the thermal structure of a subduction zone
- Explain what is being melted to produce primary magmas in subduction zones



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Magmatism at destructive plate margins

3.1 Compositions of subduction zone magmas

- The calc-alkaline trend

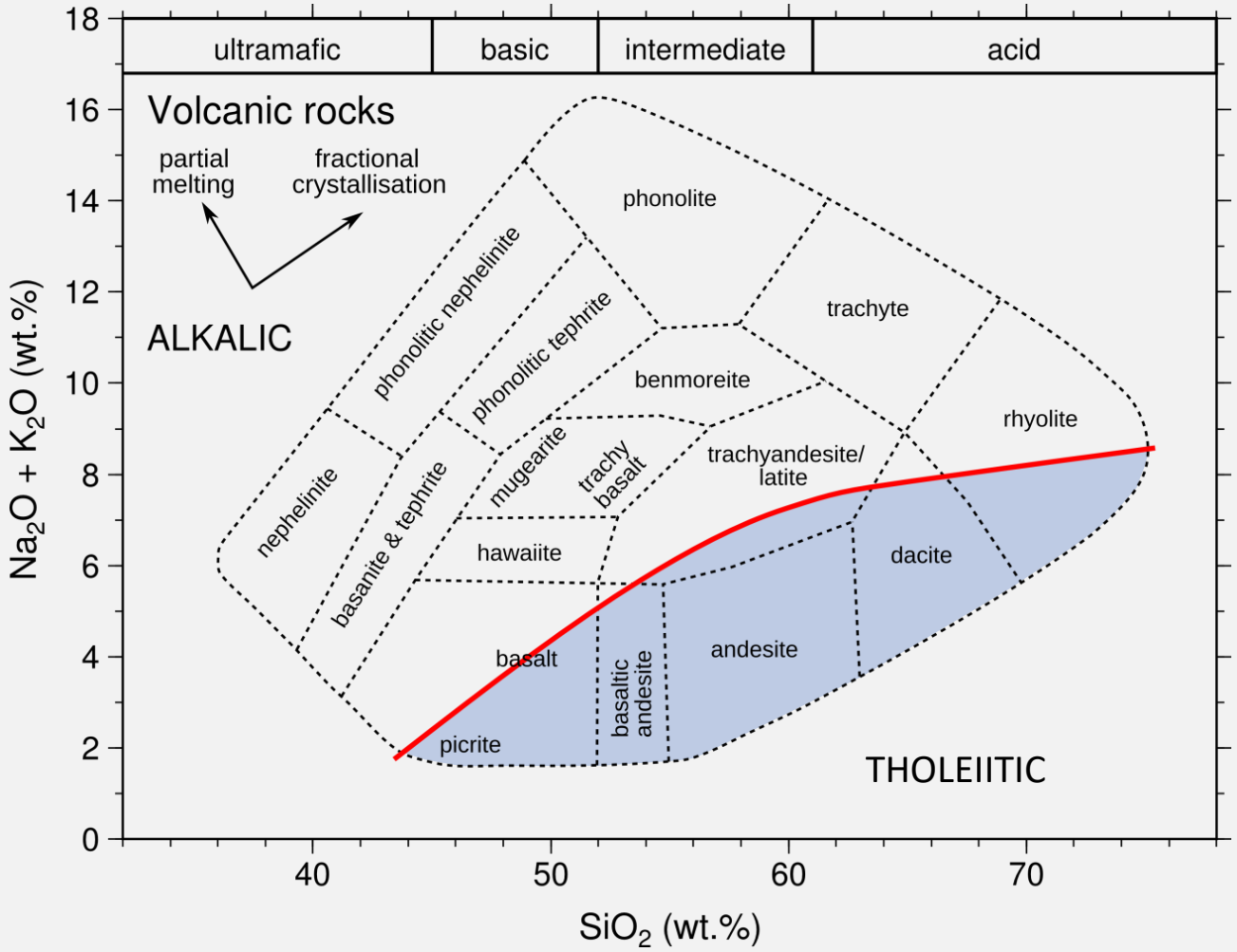
3.2 Trace element signatures of subduction zone basalts

3.3 How to make andesites, dacites and rhyolites



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Compositions of subduction zone magmas



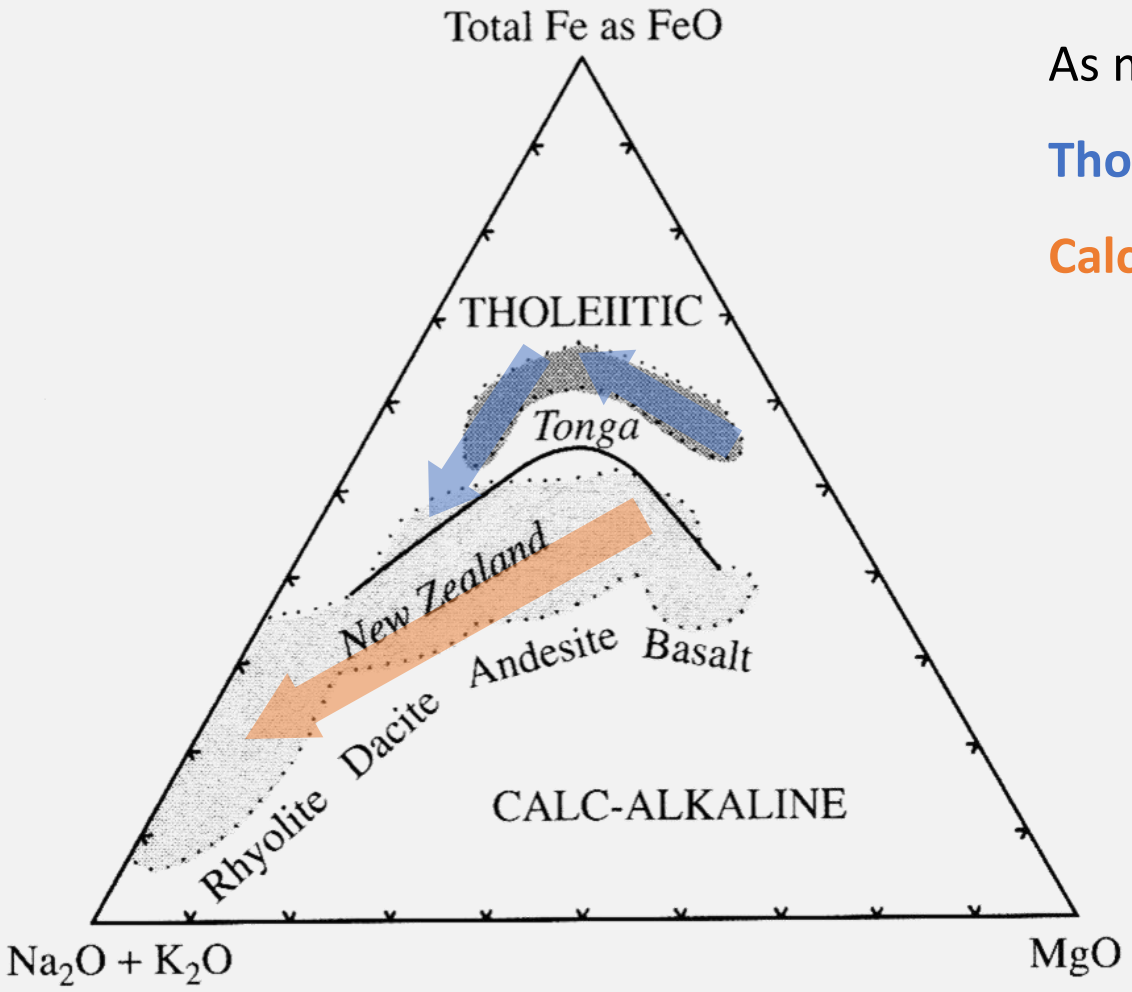
Tholeiitic suites: Typical for island arcs. Sub-alkaline, showing marked Fe enrichment. Basalts dominate, but also andesites, dacites and rhyolites.

Alkaline suites are volumetrically unimportant (<5%) at subduction zones.

Calc-alkaline suites: Common at continental arcs. Sub-alkaline, but showing little or no Fe enrichment. Andesites dominate, but also basalts, dacites and rhyolites

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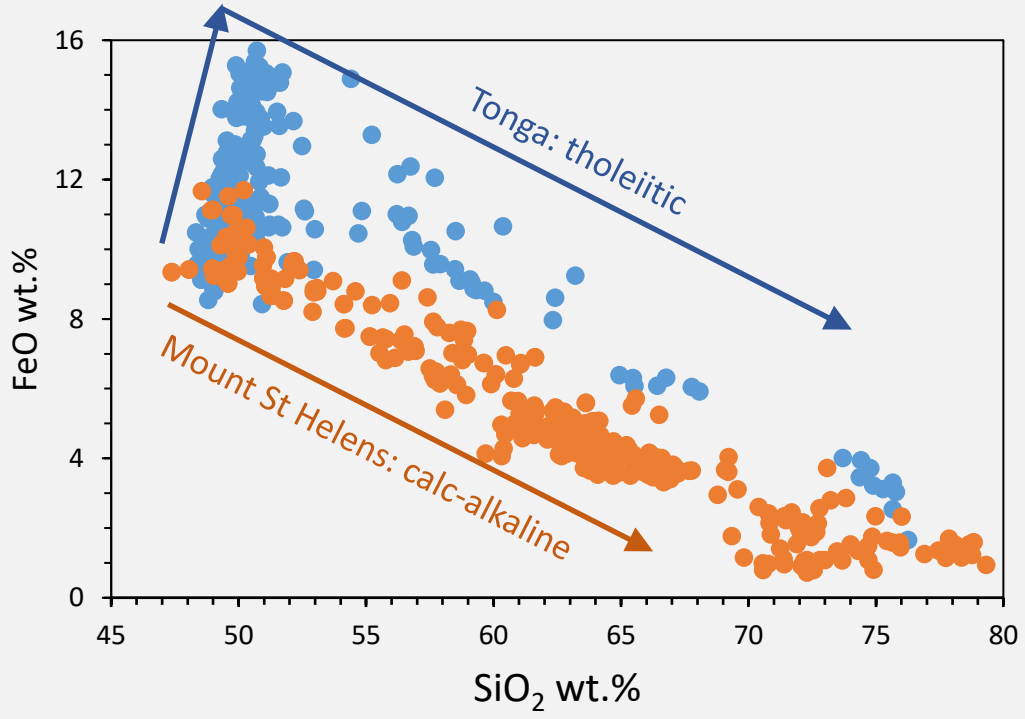
The calc-alkaline trend



As magmas evolve towards more felsic compositions:

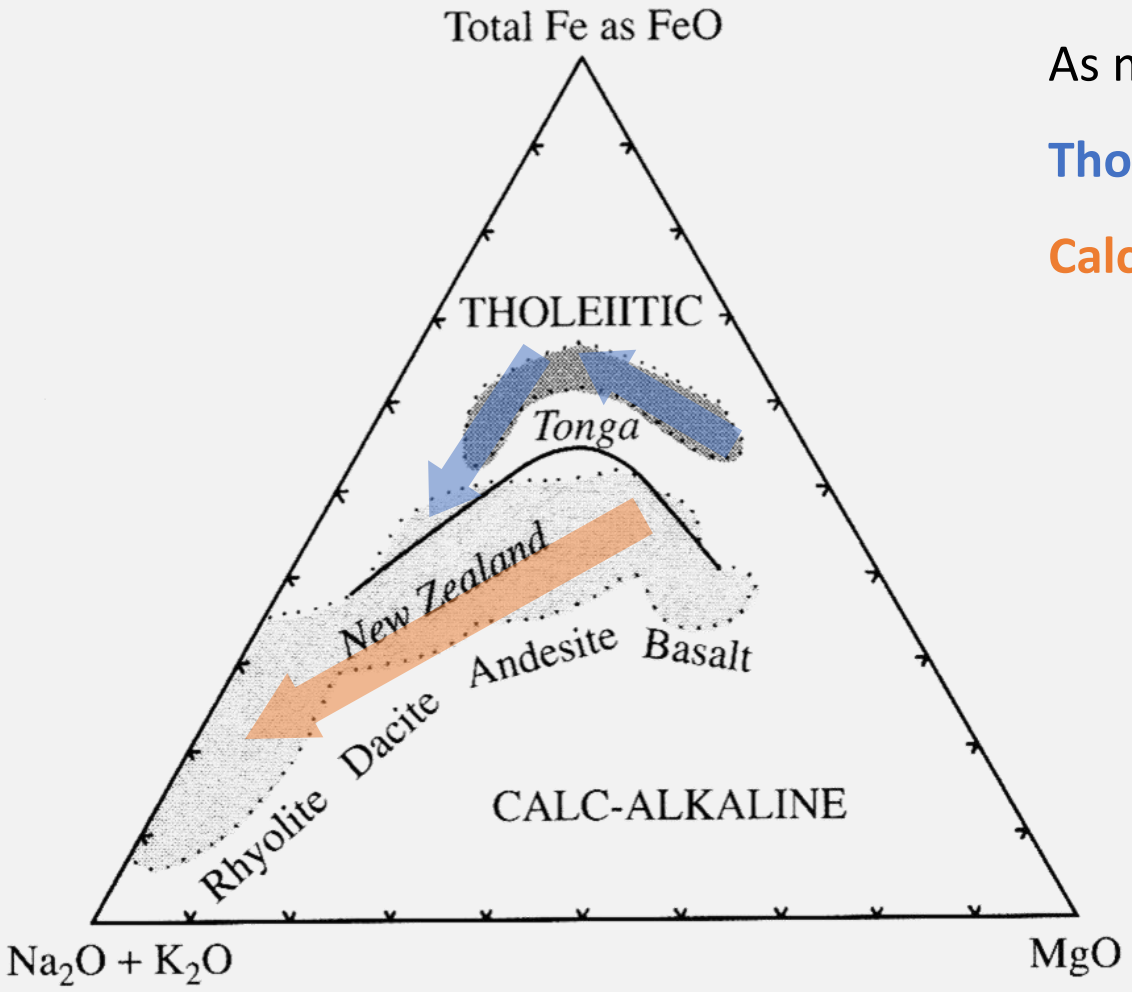
Tholeiitic fractionation trend shows initial Fe enrichment

Calc-alkaline fractionation trend shows no Fe enrichment



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The calc-alkaline trend



As magmas evolve towards more felsic compositions:

Tholeiitic fractionation trend shows initial Fe enrichment

Calc-alkaline fractionation trend shows no Fe enrichment

Some proposed explanations for the calc-alkaline trend:

- Magma mixing between basalt and rhyolite?
- Crustal assimilation?
- Suppression of plagioclase crystallization (expected for wet melts at high pressure)
- Enhanced crystallization of Fe-Ti oxides in calc-alkaline magmas, due to oxidising conditions

Read more in Gill (2010) Chapter 6 (optional)

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Magmatism at destructive plate margins

3.1 Compositions of subduction zone magmas

- The calc-alkaline trend

3.2 **Trace element signatures of subduction zone basalts**

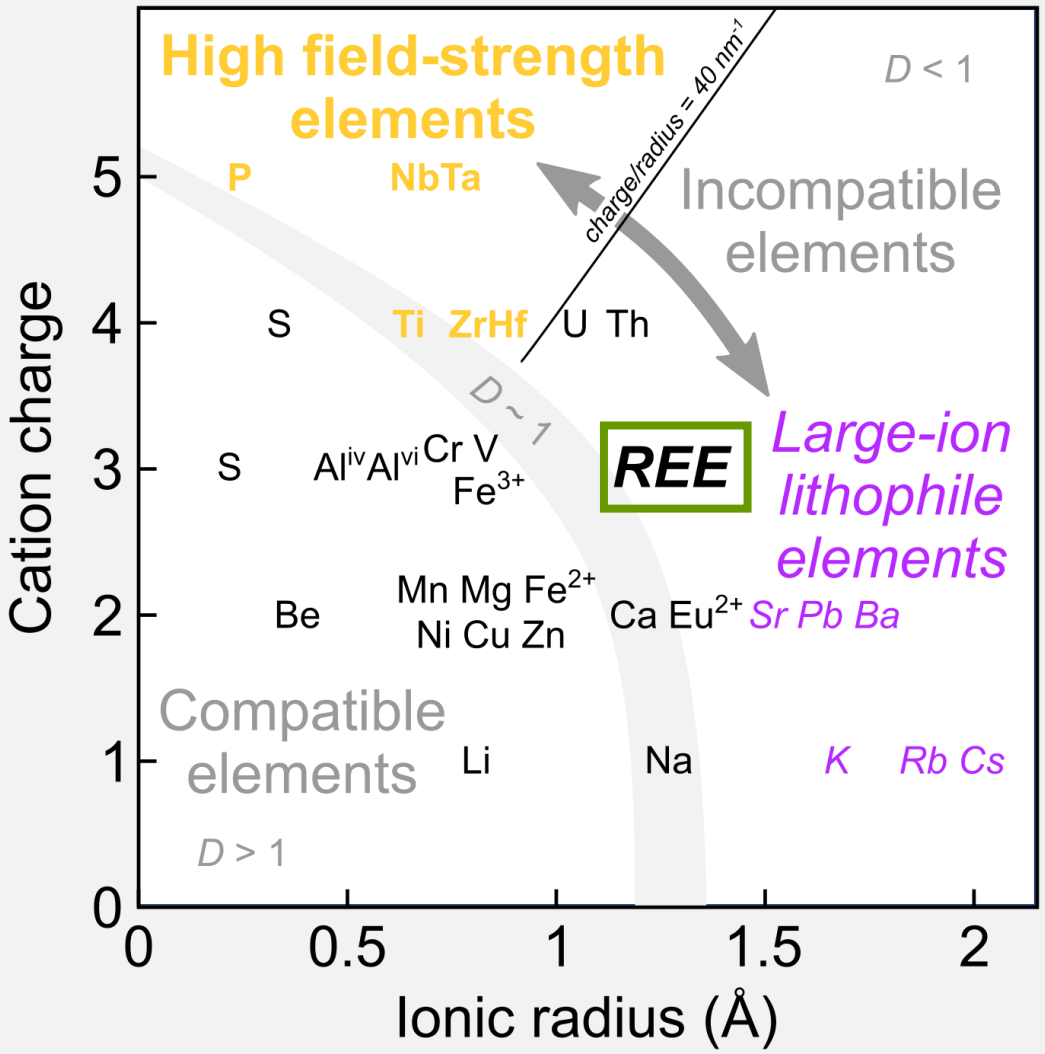
3.3 How to make andesites, dacites and rhyolites



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Recap: Types of incompatible trace element

See also EART 26201 Principles of Geochemistry, Lecture 1



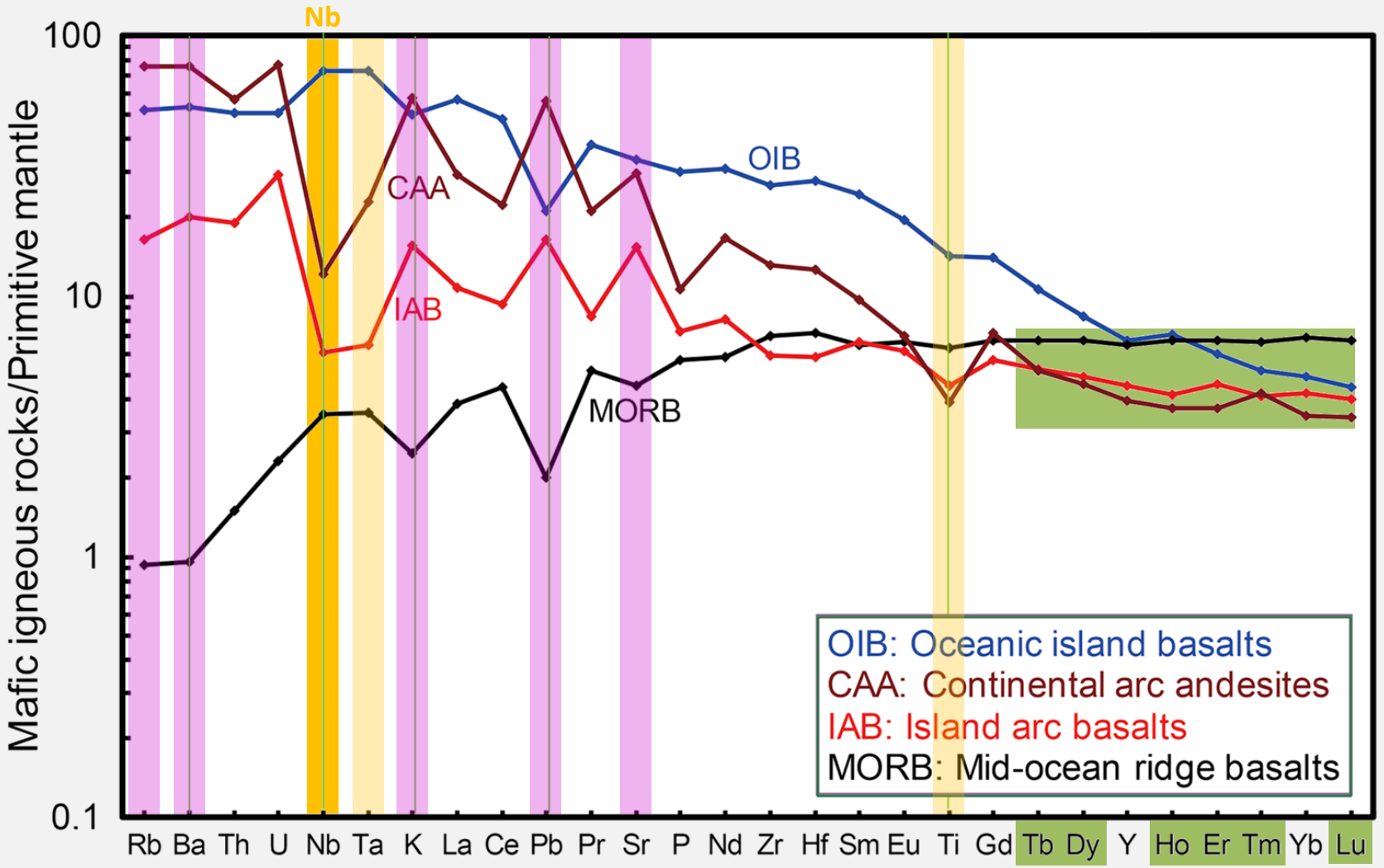
High field strength elements (HFSE)
Too highly charged to fit into crystal lattices
Largely immobile in aqueous fluids

Large ion lithophile elements (LILE)
Too large to fit into crystal lattices
Highly mobile in aqueous fluids

Rare earth elements (REE)
Show related behaviour during key geological processes like melting and crystallisation

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Trace element compositions of subduction zone magmas



Continental and island arc basalts are:

- Generally enriched in incompatible trace elements
- Enriched in **LILE**, which are transported to the mantle wedge by aqueous fluids derived from the subducting slab
- Depleted in **HFSE**, which are retained in the slab
- Strong negative **Nb** anomaly is very characteristic of subduction zone magmas
- Slope of **HREE** profile indicates whether melting was mainly of garnet or spinel peridotite

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Magmatism at destructive plate margins

3.1 Compositions of subduction zone magmas

- The calc-alkaline trend

3.2 Trace element signatures of subduction zone basalts

3.3 How to make andesites, dacites and rhyolites



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Andesite: what to look for in a thin section

Coarse-grained equivalent is diorite

Essential minerals found in <u>all</u> andesites:

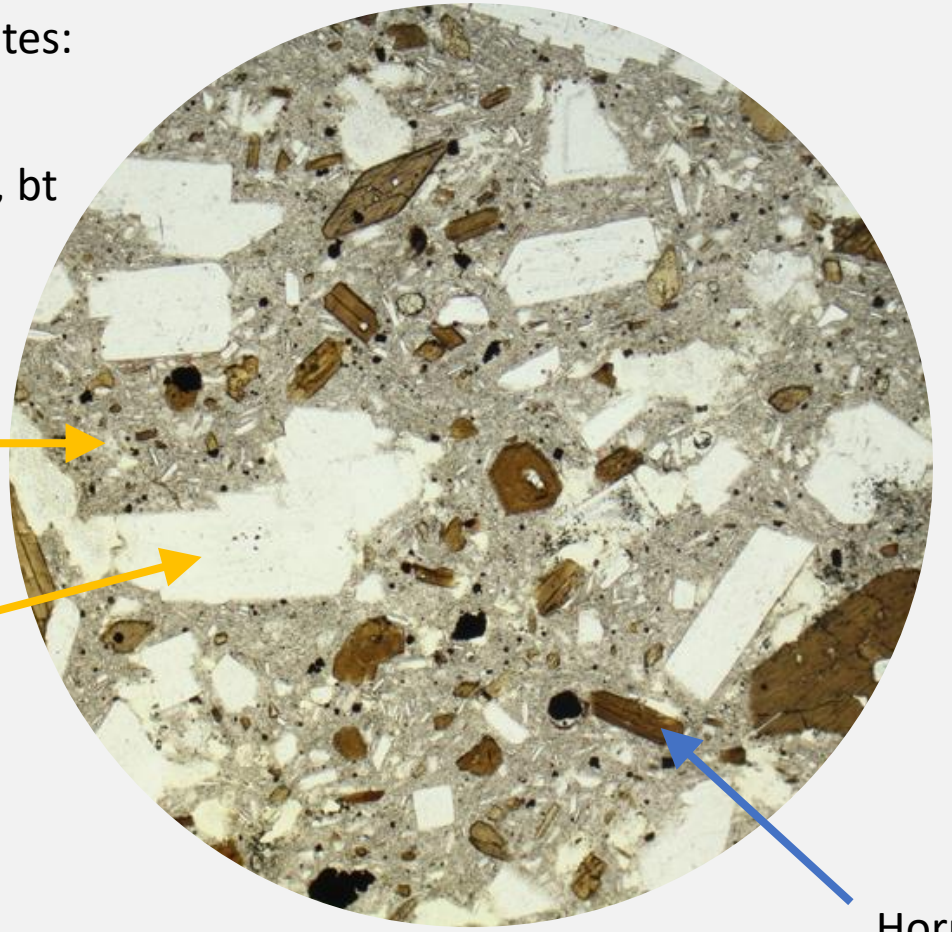
- Plagioclase (intermediate Ca)
- At least one mafic mineral: px, hbl, bt

Fine-grained groundmass contains the essential minerals

Plagioclase phenocrysts

Andesites often markedly plagioclase-phyric

Porphyritic or glomeroporphyritic texture



Hornblende phenocryst

Type minerals in andesites:

- Clinopyroxene (augite)
- Orthopyroxene
- Olivine
- Hornblende
- Biotite
- Oxides e.g. magnetite

Hydrous minerals indicate wet magma

One or more type minerals from this list may be present, but usually not all of them together

Type minerals may be present as phenocrysts and/or in the groundmass

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Dacite: what to look for in a thin section

Coarse-grained equivalent is granodiorite

Essential minerals found in <u>all</u> dacites:

- Plagioclase (high-Na)
- Alkali feldspar (sanidine)
- Quartz

plag > afs

Type minerals in dacites:

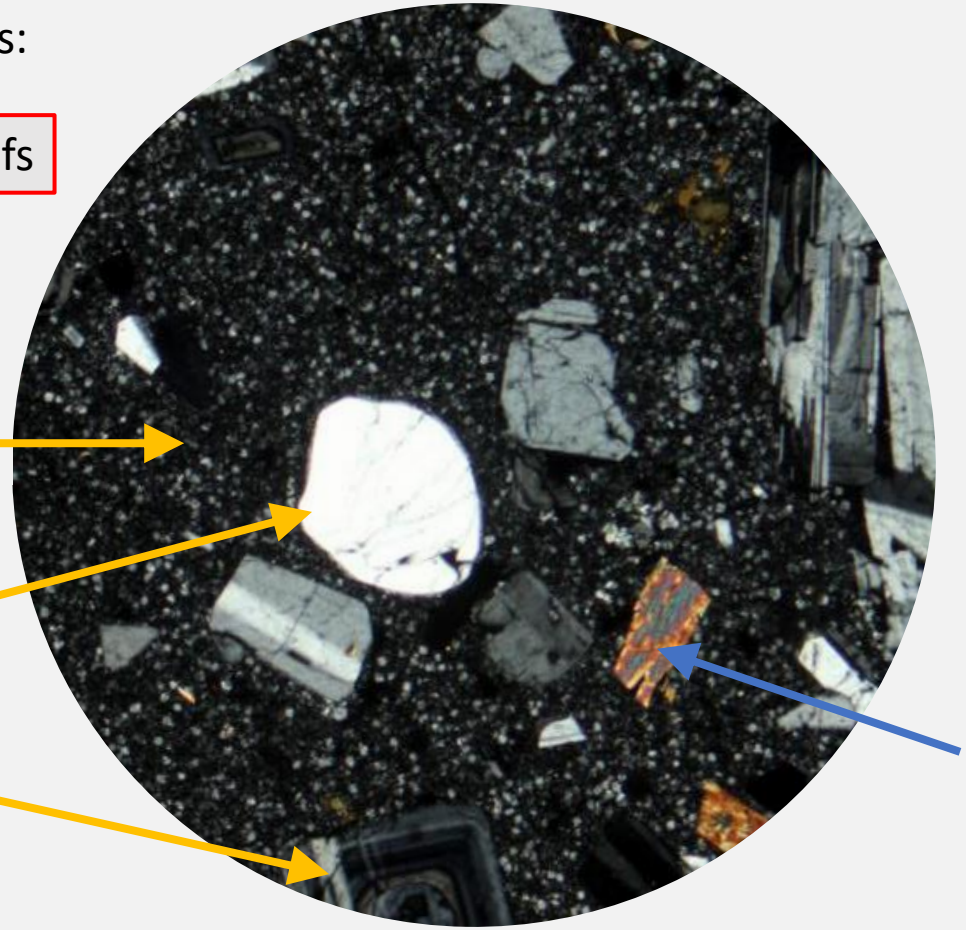
- Hornblende
- Biotite

Hydrous minerals indicate wet magma

- Pyroxene (opx or cpx)

One or more type minerals from this list may be present, but usually not all of them together

Type minerals may be present as phenocrysts and/or in the groundmass



Fine-grained groundmass contains the essential minerals

Groundmass usually crystalline

Quartz phenocryst

Plagioclase phenocryst

Dacites often quartz- and plagioclase-phyric

Hornblende phenocryst

Porphyritic or glomeroporphyritic texture

Mafic minerals less abundant than in andesite

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Rhyolite: what to look for in a thin section

Coarse-grained equivalent is granite

Essential minerals found in <u>all</u> rhyolites:

- Plagioclase (high-Na)
- Alkali feldspar (sanidine)
- Quartz

afs > plag

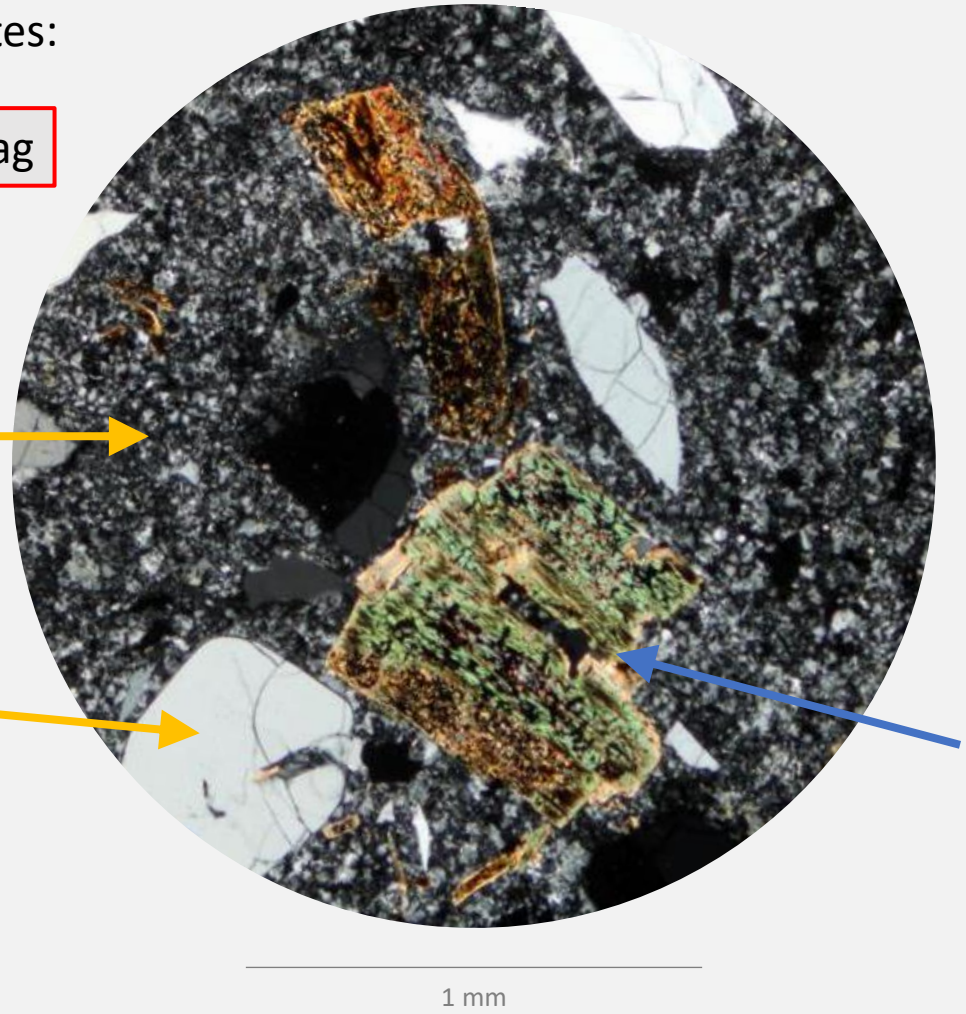
Fine-grained groundmass contains the essential minerals

Groundmass is often glassy

Quartz phenocryst

Rhyolites often quartz- and sanidine-phyric

May have banded texture



Type minerals in rhyolites:

- Hornblende
- Biotite

Hydrous minerals indicate wet magma

- Pyroxene (opx or cpx)

One or more type minerals from this list may be present, but usually not all of them together

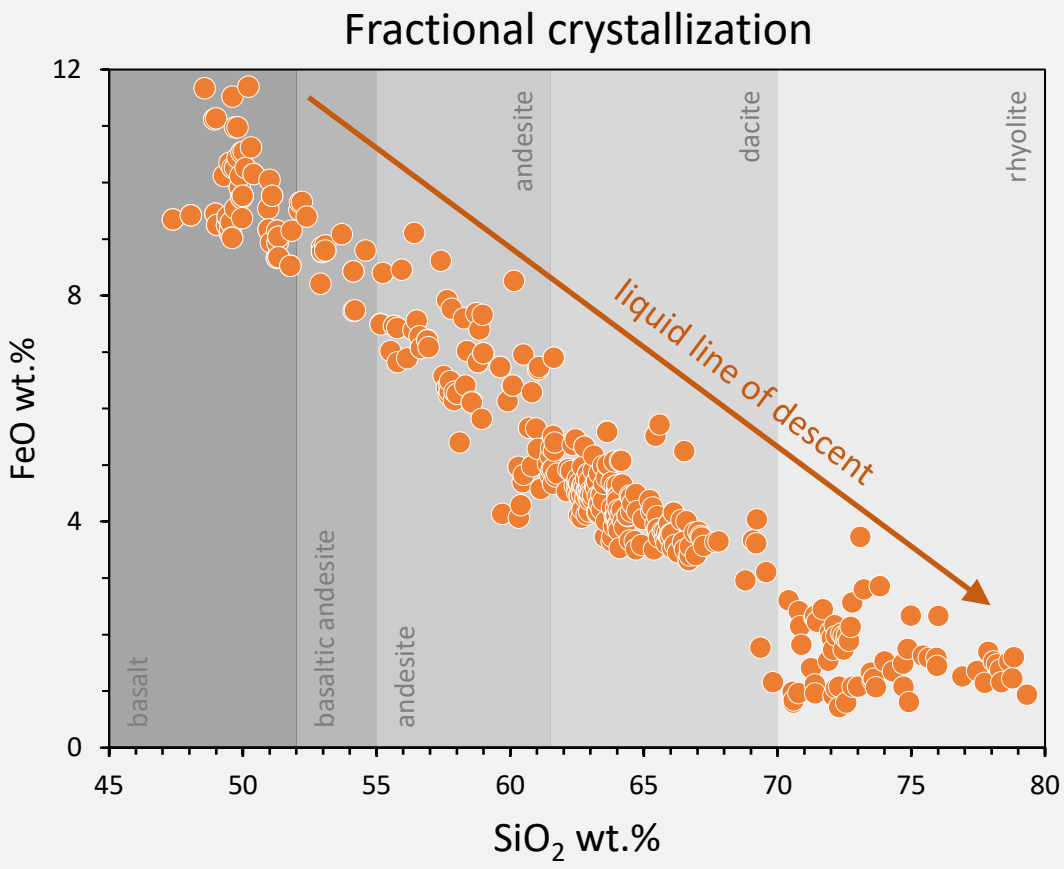
Type minerals may be present as phenocrysts and/or in the groundmass

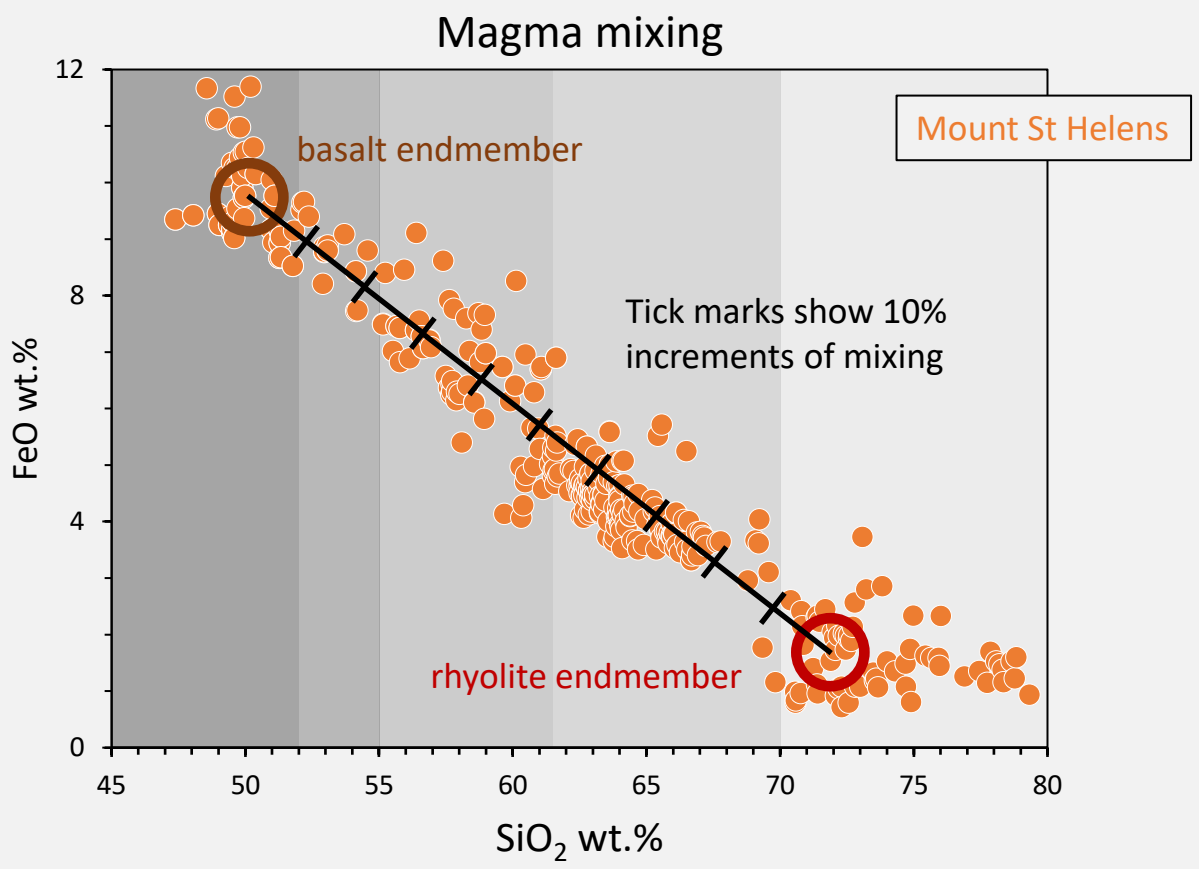
Biotite phenocryst

More on pyroclastic rocks in Week 10

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Intermediate and evolved magmas: fractional crystallization or something else?

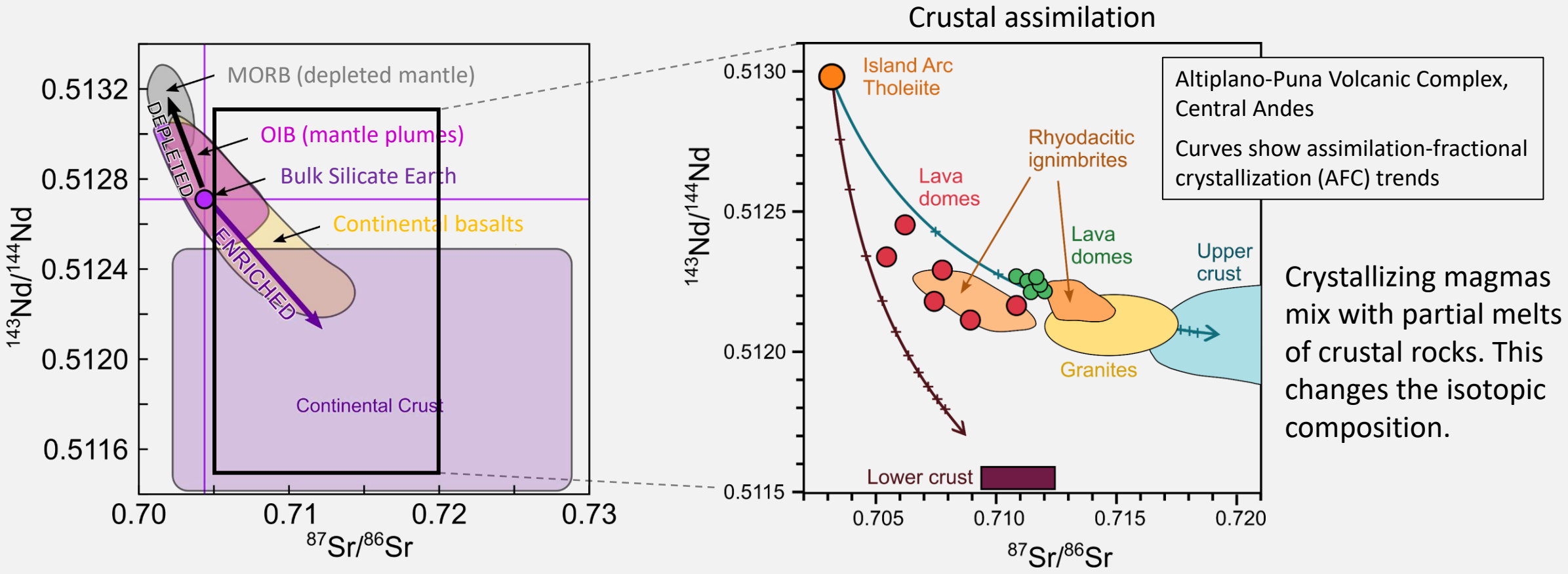




Straight-line segments on single Harker plots don't provide the information needed to distinguish between crystallization and mixing.

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Intermediate and evolved magmas: fractional crystallization or something else?



Altiplano-Puna Volcanic Complex, Central Andes

Curves show assimilation-fractional crystallization (AFC) trends

Crystallizing magmas mix with partial melts of crustal rocks. This changes the isotopic composition.

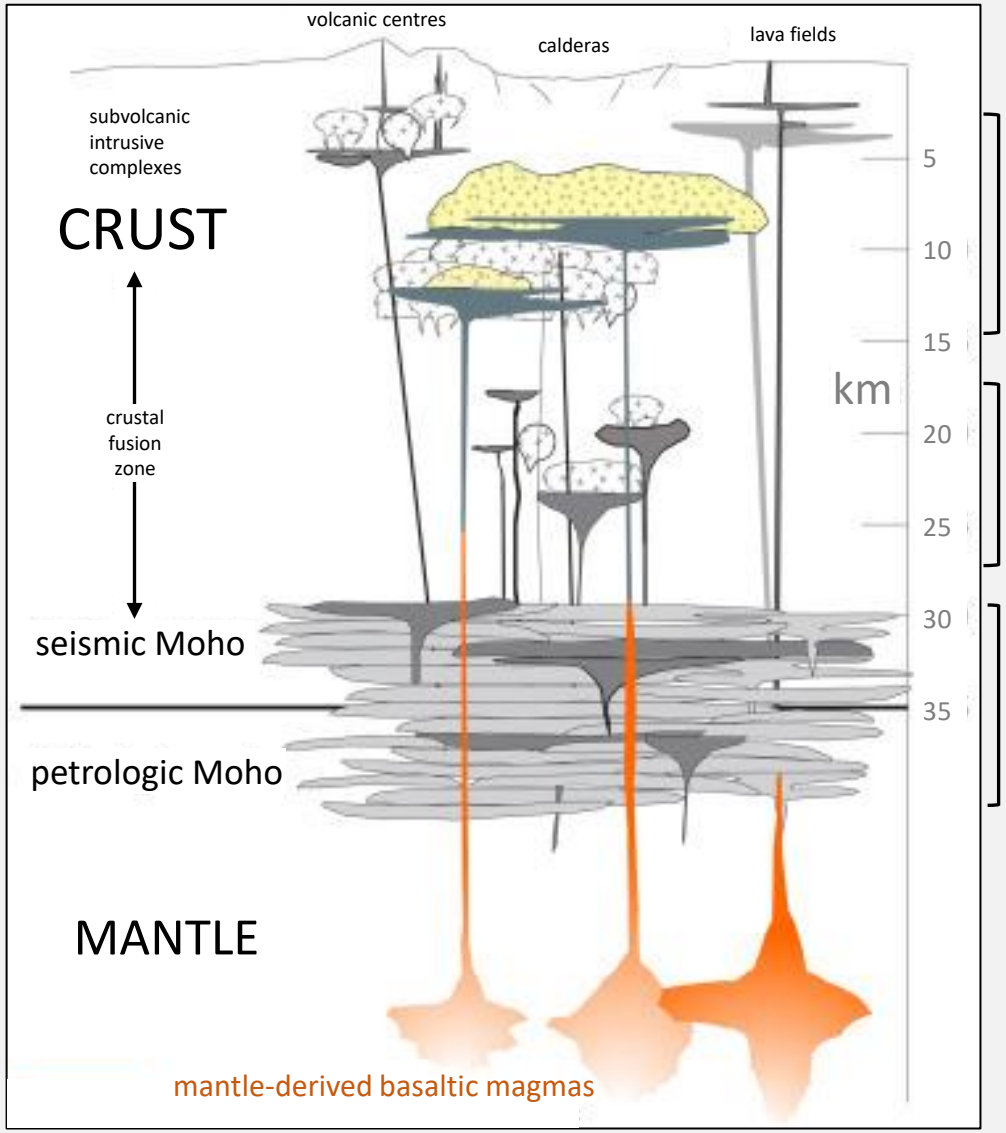
Isotopes are not fractionated (much) by high-T processes like fractional crystallization.

Crust-like isotope ratios in intermediate and evolved magmas point to crustal assimilation.
Contintental arc magmas usually have isotopic signatures trending towards average continental crust.

See EART 26201 Principles of Geochemistry, Lecture 4

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Assimilation and fractional crystallization can happen throughout the crust



Shallow plumbing system: Upper crustal assimilation + fractional crystallization

Deeper plumbing system: Lower crustal assimilation + fractional crystallization

Underplating, fractional crystallization, magma mixing, partial melting

Large evolved silicic magma bodies formed by large-scale crustal assimilation

Basaltic magmas begin to induce partial melting of crust, which produces evolved magmas

Mainly basaltic magmas

Fractional crystallization and crustal assimilation can happen anywhere in the magmatic system. Mafic recharge and magma mixing add to the compositional and mineralogical diversity.

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Magmatism at destructive plate margins

You should now be able to:

- Describe the compositional diversity of magmas erupted at subduction zone volcanoes
- Outline the principal differences between tholeiitic and calc-alkaline suites
- Describe the characteristic trace element signatures of subduction zone magmas, and explain why these signatures are observed
- Discuss the role of crustal assimilation in generating andesitic, dacitic and rhyolitic magmas



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Magmatism at destructive plate margins

4.1 Compositional zoning in phenocrysts

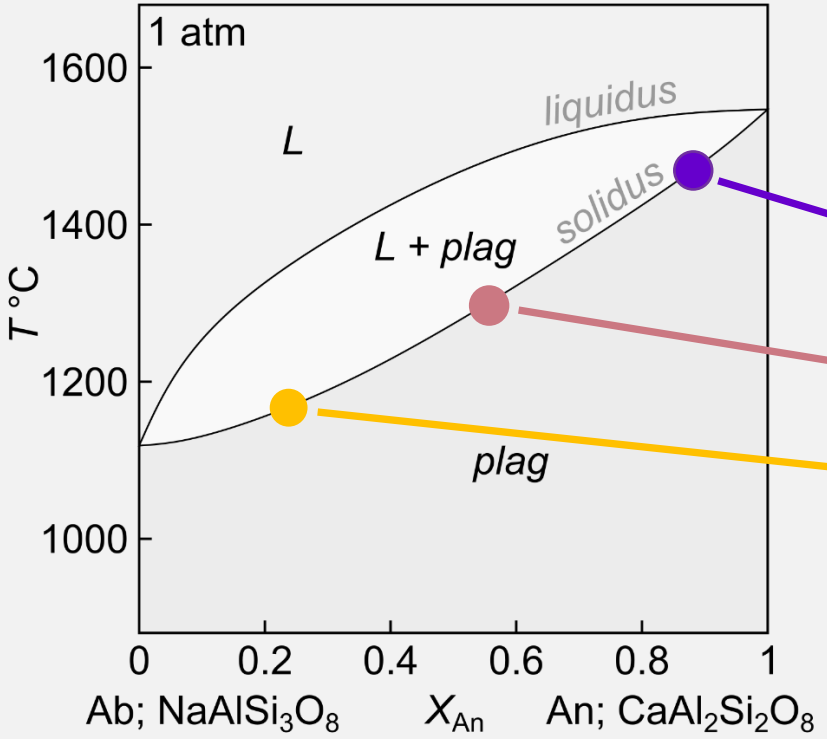
4.2 Chemical disequilibrium in subduction zone magmas

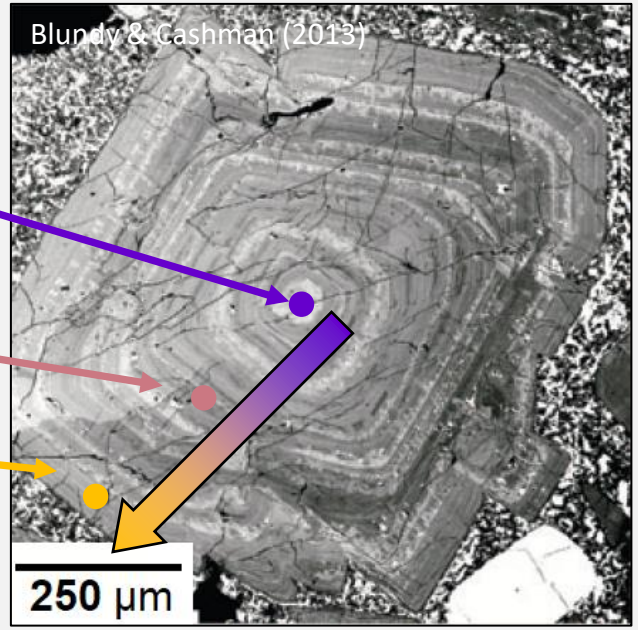


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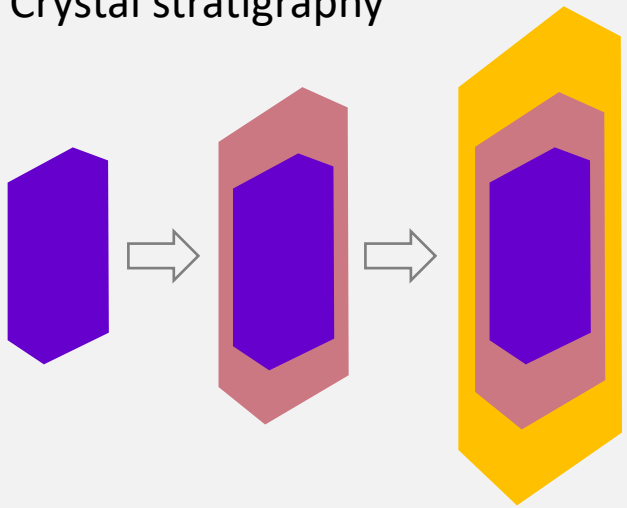
Normal zoning in magmatic phenocrysts

See also EART 26201 Principles of Geochemistry, Lecture 5





- Crystals grow in concentric zones
- Zones record P, T, X conditions at different stages of crystal growth
- Crystal stratigraphy



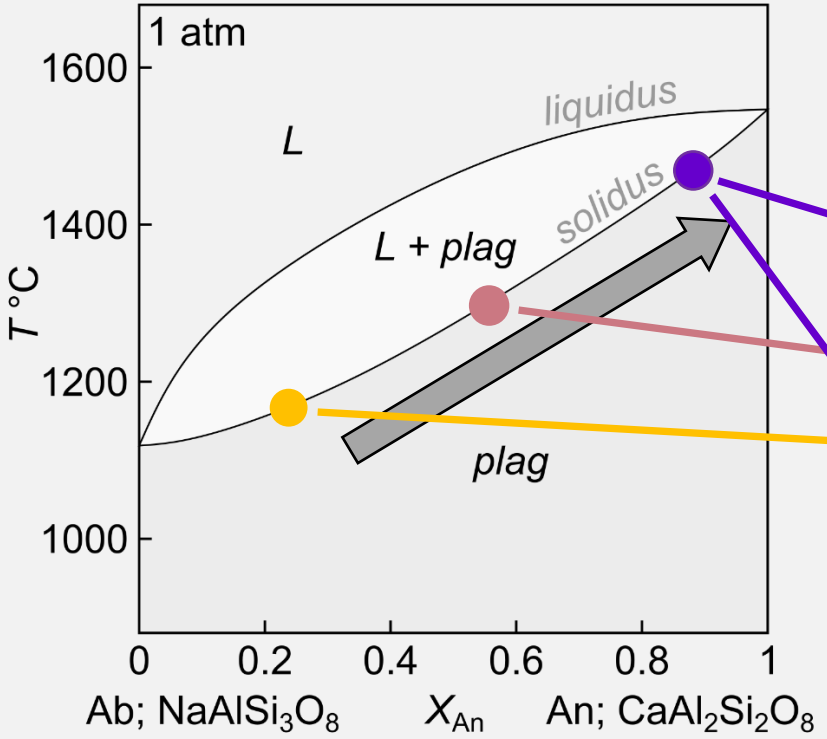
Normal zoning generated due to continuous change of magma T and X during simple magma crystallization.

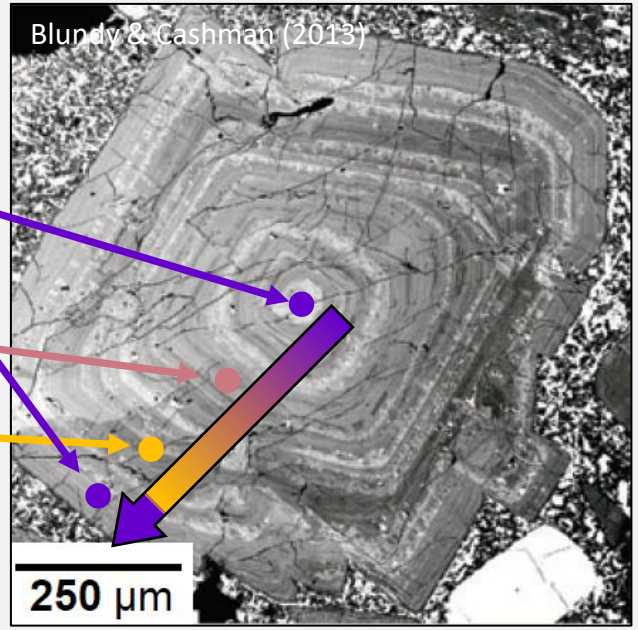
Crystal zones are preserved because the kinetics of cation diffusion are slower than the rate of crystal growth.

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Reverse zoning in magmatic phenocrysts

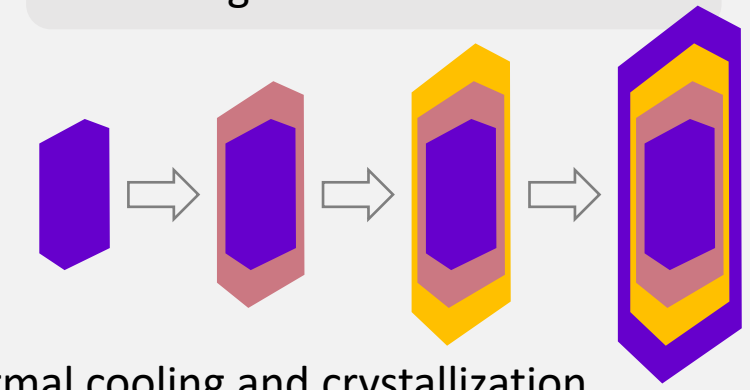
See also EART 26201 Principles of Geochemistry, Lecture 5





- Crystals grow in concentric zones
- Zones record P, T, X conditions at different stages of crystal growth
- Crystal stratigraphy

Mafic recharge: injection of new, hot magma into the chamber



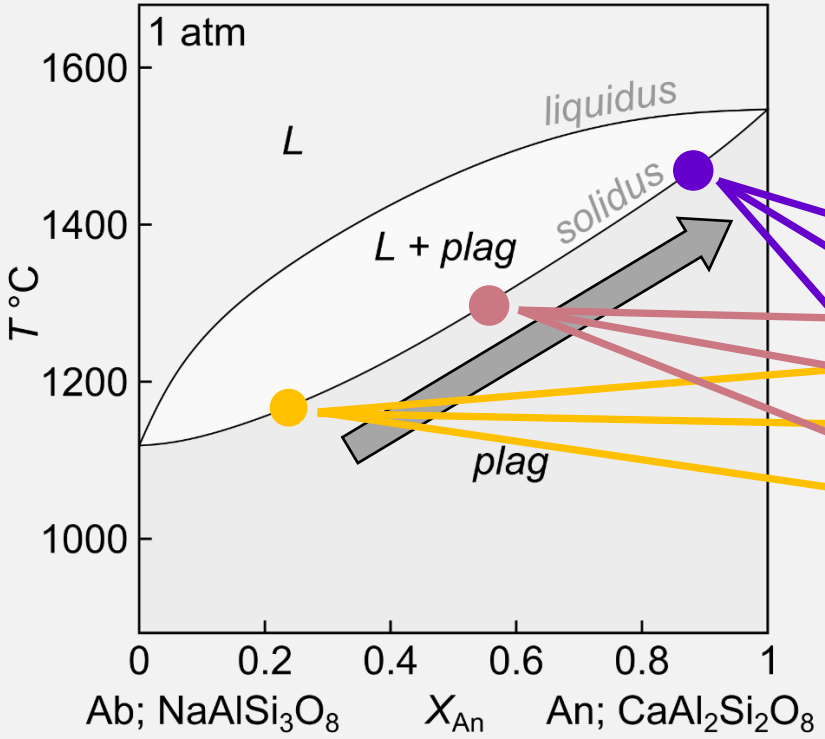
Reverse zoning requires magma T and X to change in the opposite sense to normal cooling and crystallization

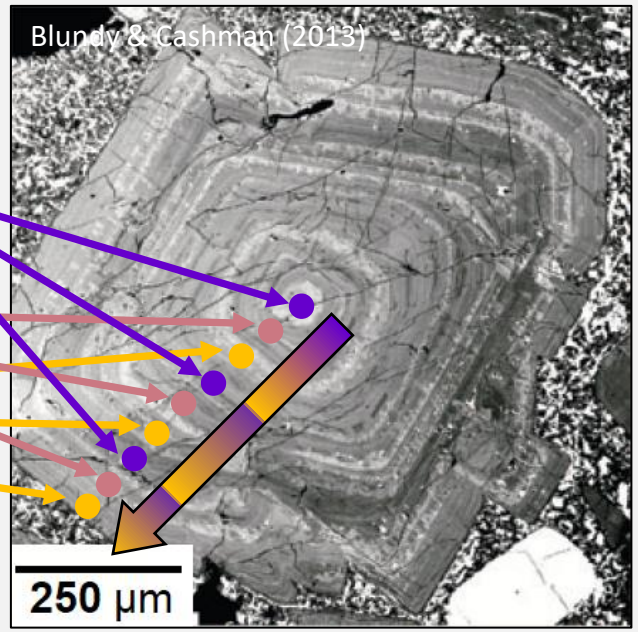
Example: mafic recharge of the magma chamber (changes X and increases T)

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Oscillatory zoning in magmatic phenocrysts

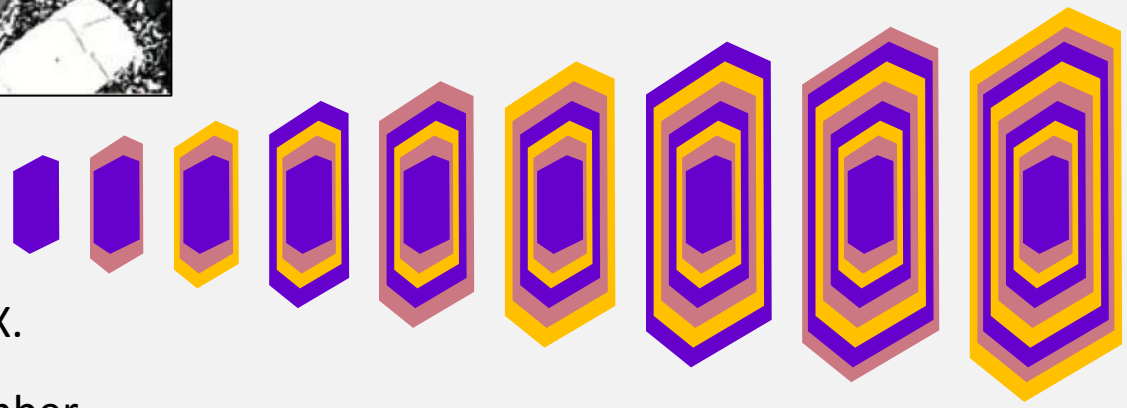
See also EART 26201 Principles of Geochemistry, Lecture 5





- Crystals grow in concentric zones
- Zones record P, T, X conditions at different stages of crystal growth
- Crystal stratigraphy

Mafic recharge: injection of new, hot magma into the chamber

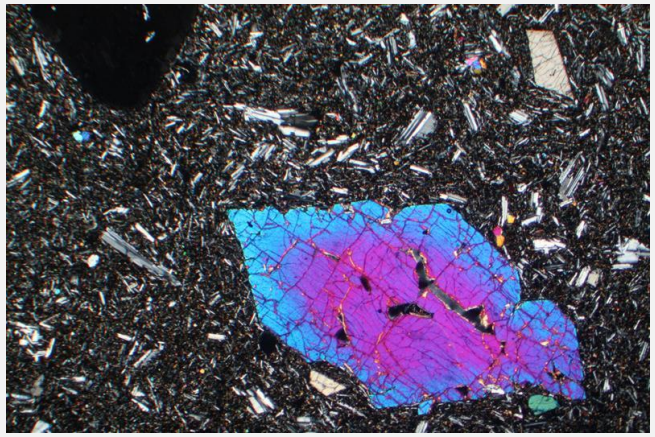


Oscillatory zoning requires repeated changes to magma T and/or X.

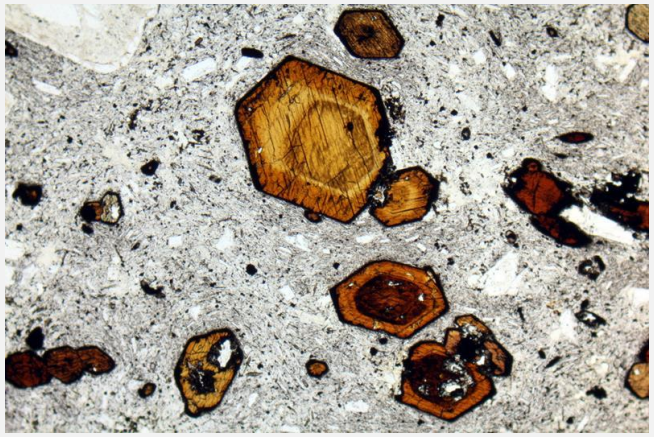
Example: repeated episodes of mafic recharge in the magma chamber

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Zoning can occur in any mineral that has solid solution



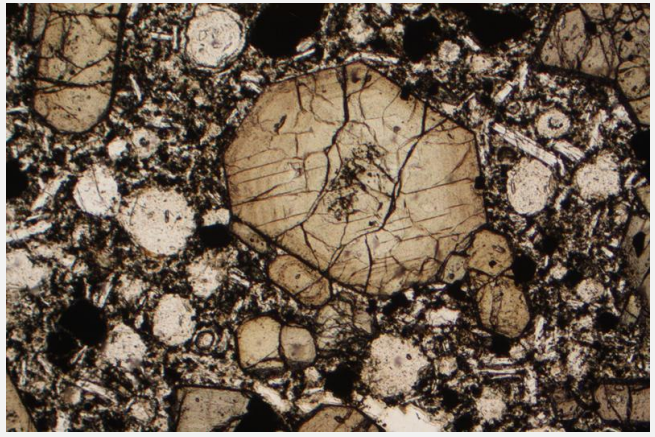
Zoned olivine in a basalt



Zoned brown hornblende in an andesite



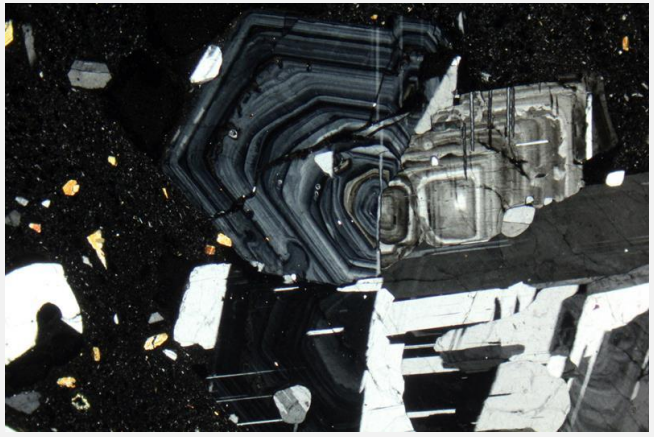
Zoned green hornblende in a rhyolite



Zoned augite (cpx) in an alkali basalt



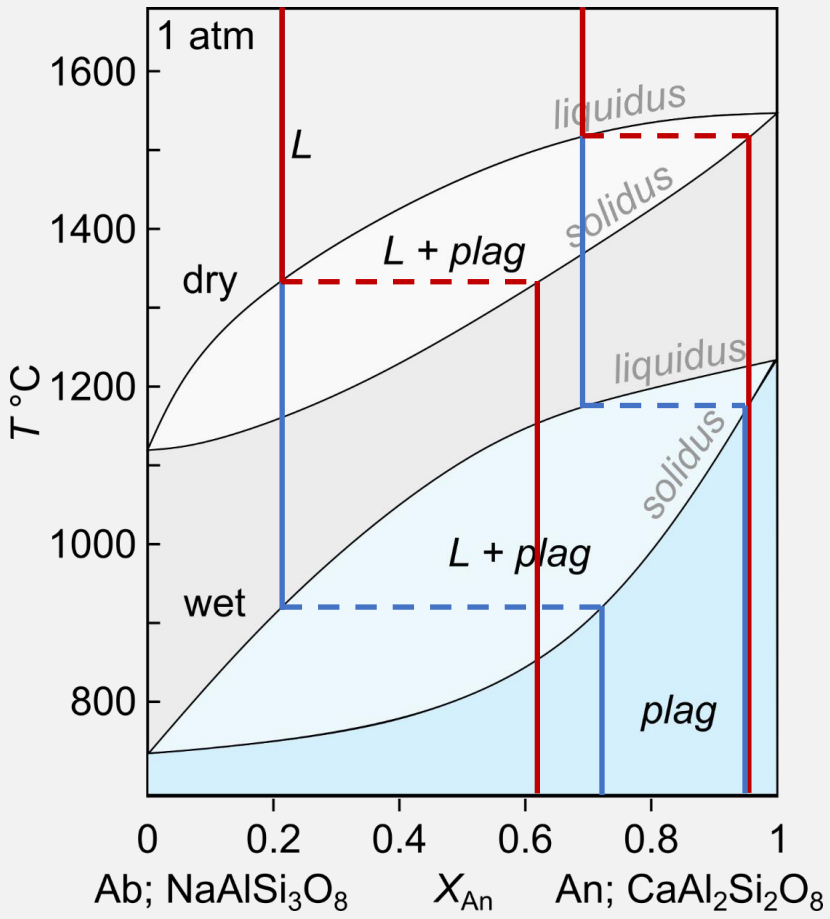
Zoned plagioclase in a diorite



Zoned plagioclase in a rhyolite

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Plagioclase phase diagram changes depending on the magmatic water content



Mafic magma, XAn ~0.7

Dry: Crystallization starts at ~1500 °C; An~90 plagioclase

Wet: Crystallization starts at ~1350 °C; An~90 plagioclase

Evolved magma, XAn ~0.2

Dry: Crystallization starts at ~1300 °C; An~60 plagioclase

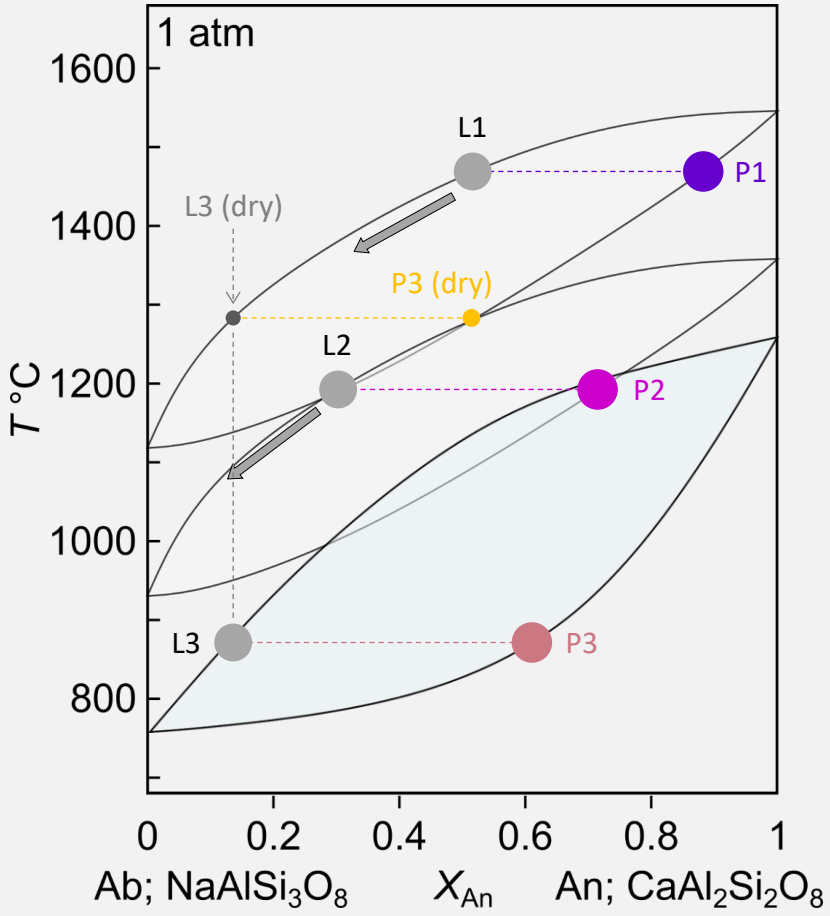
Wet: Crystallization starts at ~900 °C; An~70 plagioclase

For a fixed melt XAn:

- Adding H_2O depresses the plagioclase solidus; crystallization is suppressed until lower temperature
- Wetter melts crystallize plagioclase with more anorthitic compositions

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Plagioclase phase diagram changes depending on the magmatic water content



Liquid (melt) L1 crystallizes plagioclase P1

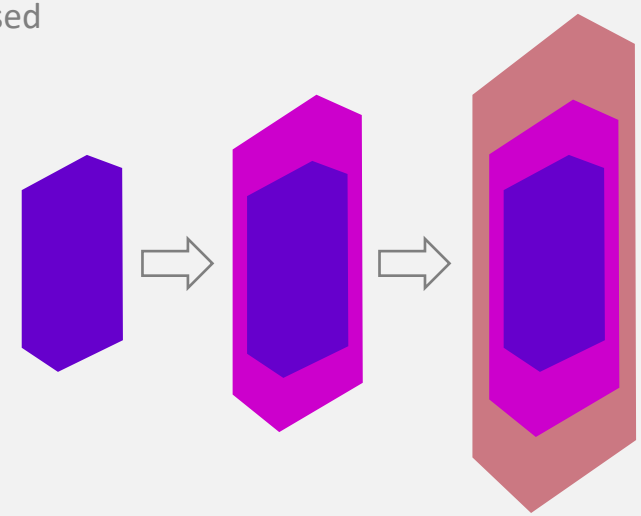
H_2O is an incompatible species, so the evolving liquid becomes more H_2O-rich.

The plagioclase liquidus and solidus are gradually depressed

Liquid (melt) L2 crystallizes plagioclase P2

Evolving liquid becomes more H_2O-rich.

Plagioclase liquidus and solidus are further depressed



Liquid (melt) L3 crystallizes plagioclase P3

Increasing the melt H_2O content at a fixed melt XAn pushes the crystallizing plagioclase to more anorthitic compositions

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Magmatism at destructive plate margins

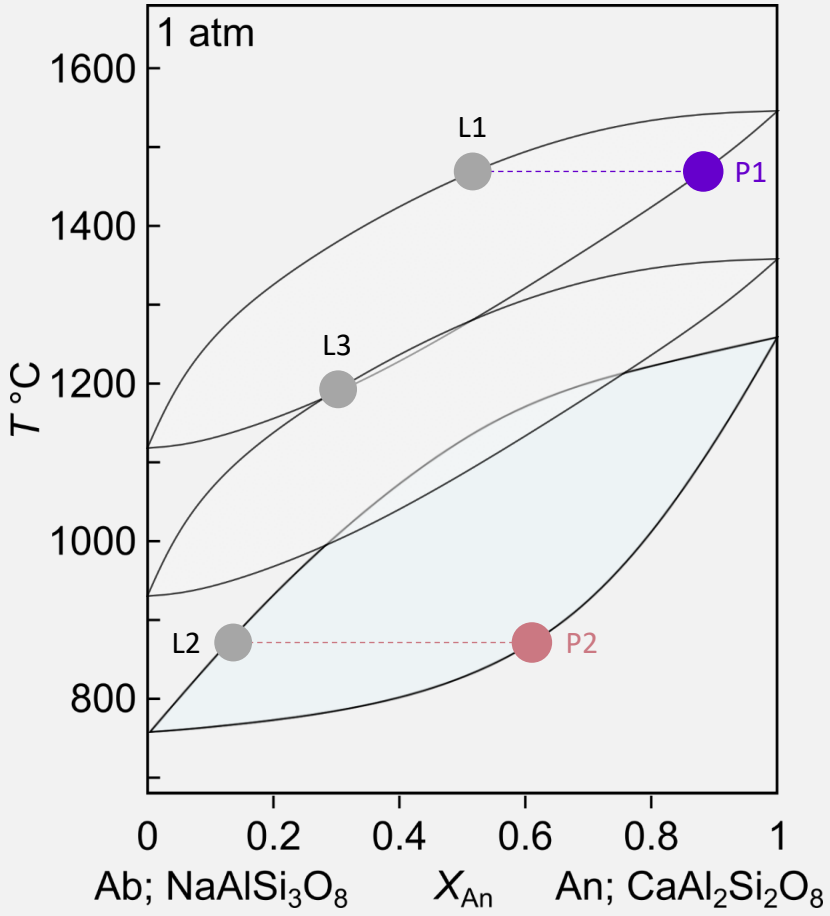
4.1 Compositional zoning in phenocrysts

4.2 Chemical disequilibrium in subduction zone magmas



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What happens when magma mixing pushes crystals out of equilibrium with the liquid?



Liquid (melt) L1 crystallizes plagioclase P1

Mafic recharge: liquid L1 and plagioclase P1 are injected into a magma chamber containing a cooler, wetter magma with liquid L2 and plagioclase P2

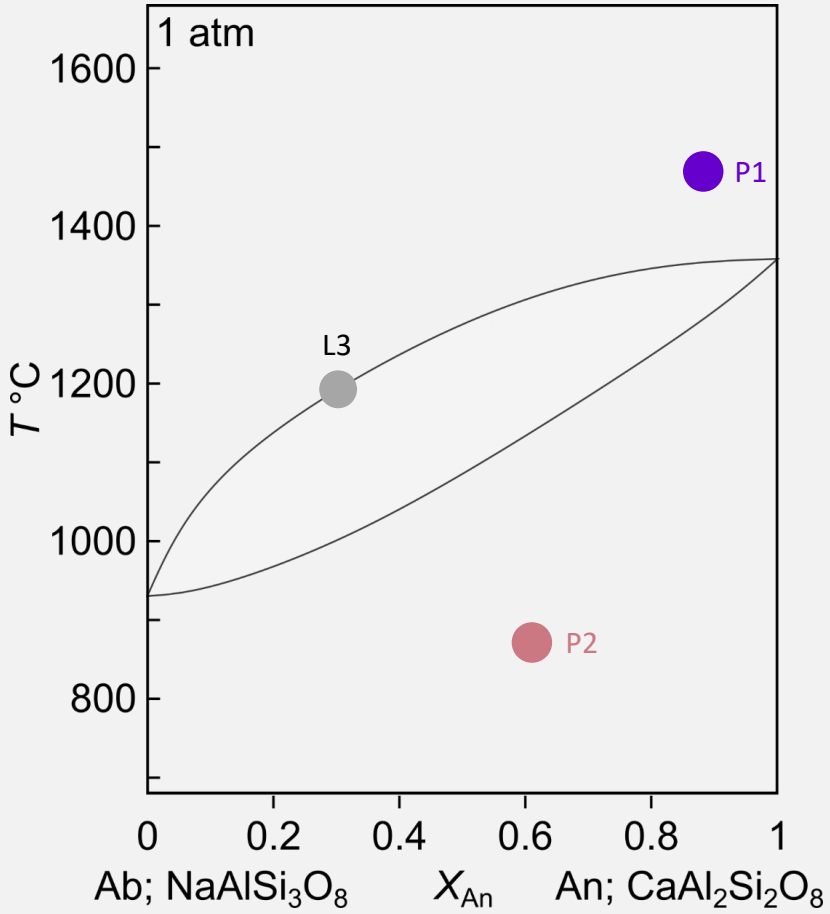
Magma mixing produces liquid L3, which contains crystals P1 and P2

What will happen to plagioclase P1?

What will happen to plagioclase P2?

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What happens when magma mixing pushes crystals out of equilibrium with the liquid?



What will happen to plagioclase P1?



What will happen to plagioclase P2?

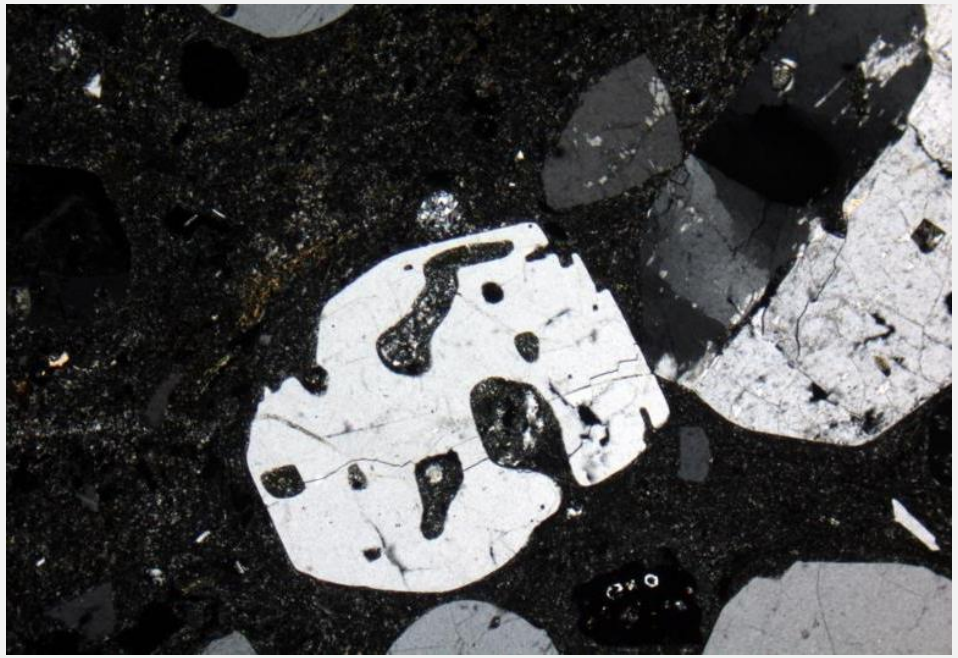


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Chemical disequilibrium can cause magmatic corrosion

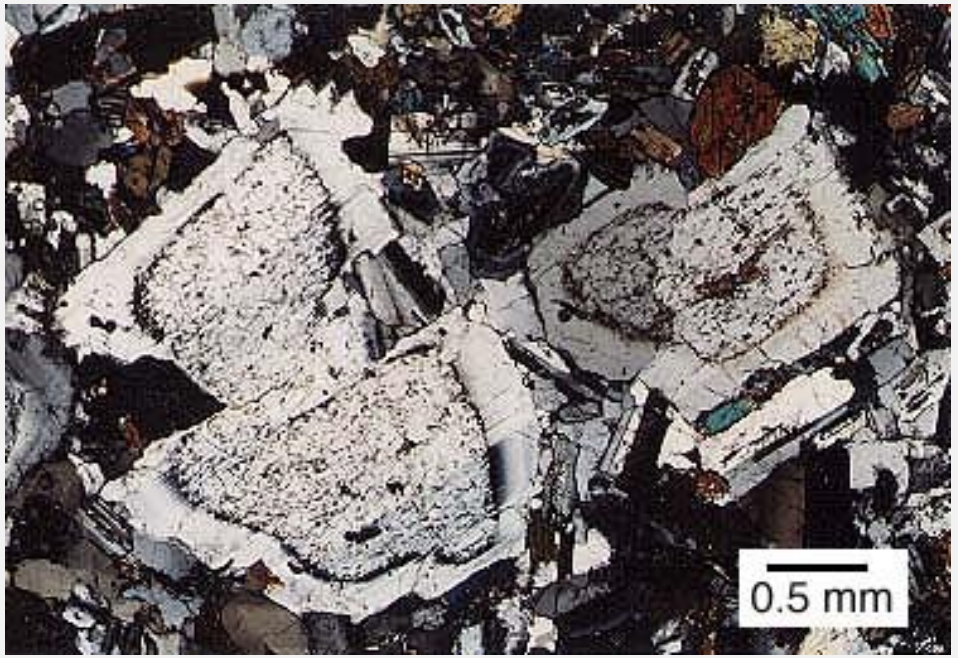
Changed magmatic conditions can cause a previously stable crystal to become chemically unstable with respect to the melt. The mineral begins resorbing (dissolving) into the melt.

Embayments



Forms by resorption without new growth. Common in quartz

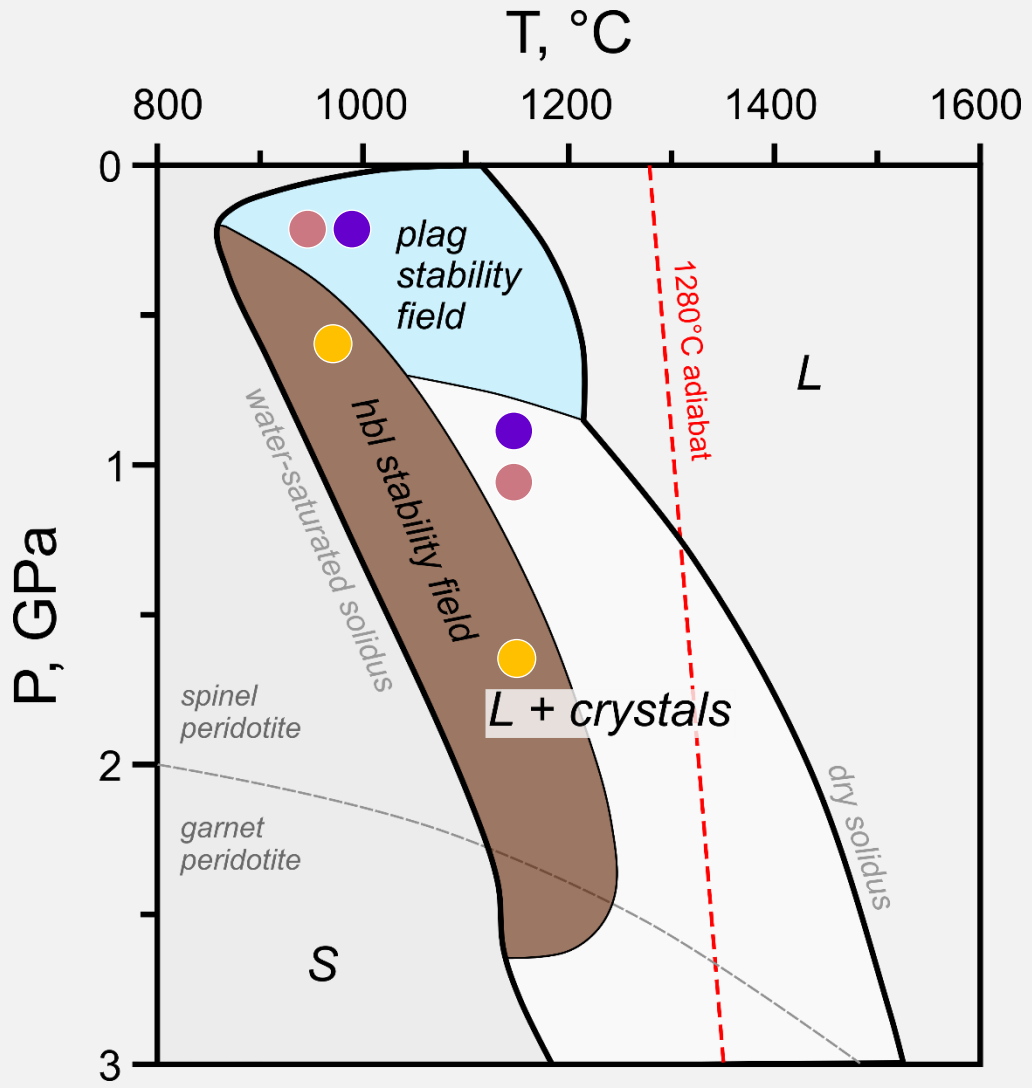
Sieve texture

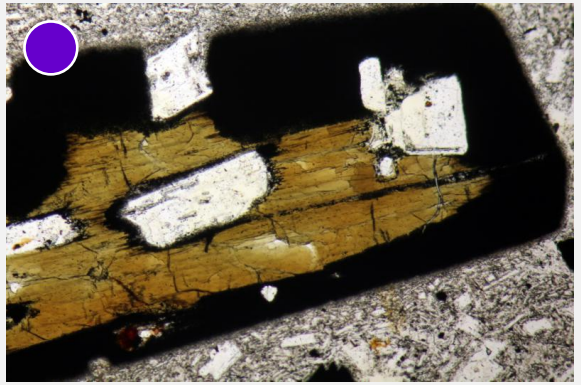


Forms by resorption followed by growth of new zone(s) once equilibrium is restored. Common in plagioclase

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Chemical disequilibrium can cause reactions between minerals and melt

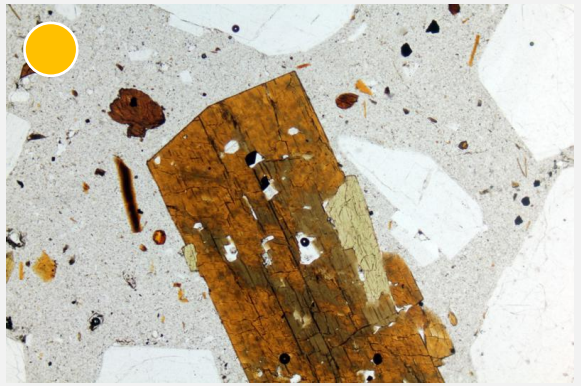




Magma spends long time in storage outside the hbl stability field: thick hbl breakdown rim



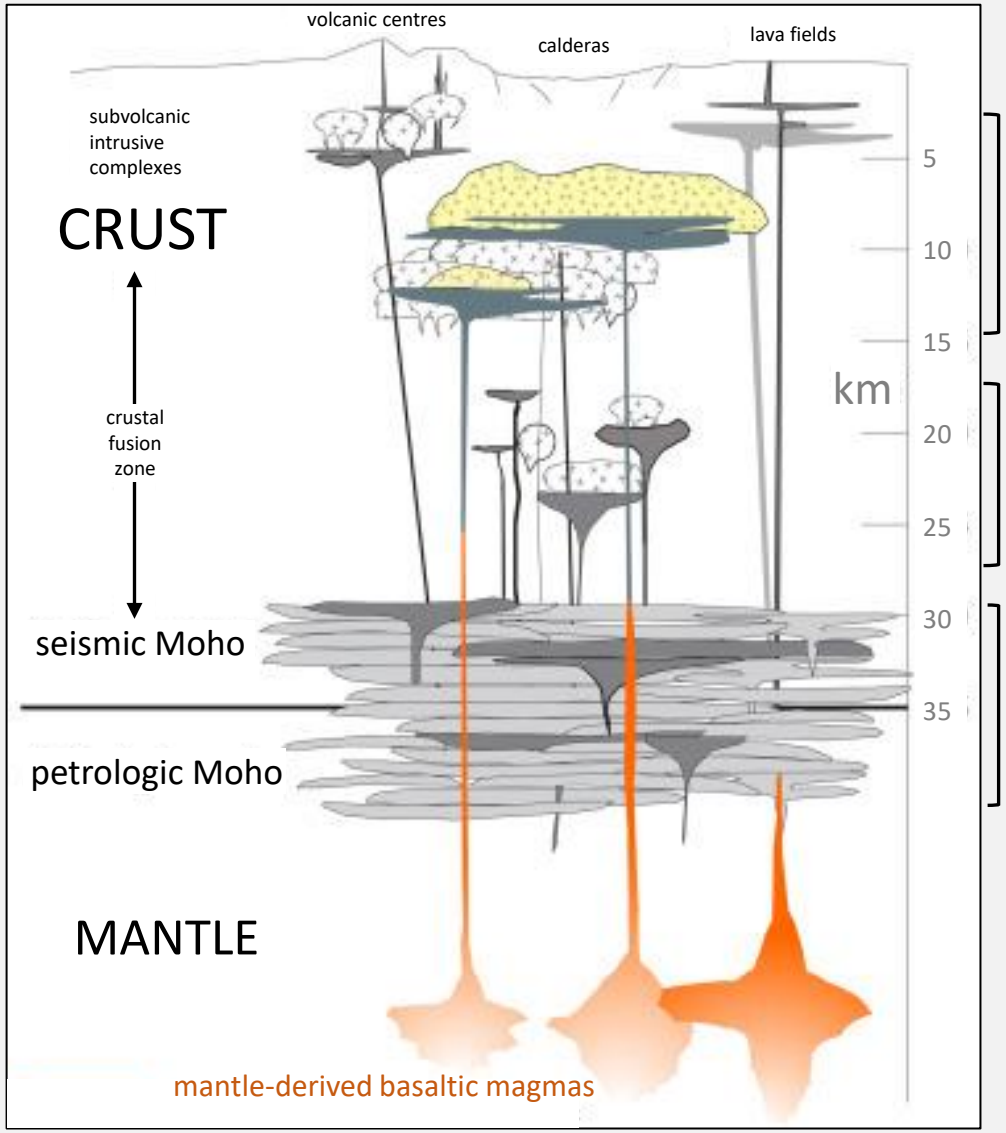
Magma spends short time outside the hbl stability field: thin hbl breakdown rim



Magma ascends directly from hbl stability field: no hbl breakdown reaction rim.

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Subduction zone magmas are chemically and texturally complex



Shallow plumbing system: Upper crustal assimilation + fractional crystallization

Large evolved silicic magma bodies formed by large-scale crustal assimilation

Deeper plumbing system: Lower crustal assimilation + fractional crystallization

Basaltic magmas begin to induce partial melting of crust, which produces evolved magmas

Underplating, fractional crystallization, magma mixing, partial melting

Mainly basaltic magmas

Lots of locations where magmas can crystallize, degas, and mix. Melts and crystals are frequently moved out of, and back into, equilibrium.

Chemical and textural indicators of disequilibrium can help petrologists interpret complex magmatic histories.

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Magmatism at destructive plate margins

You should now be able to:

- Explain the magmatic processes that cause different types of compositional zoning in phenocrysts
- Discuss the concept of chemical disequilibrium and the mineral textures this can produce
- Explain why subduction zone magmas have particularly complex magmatic histories, and discuss how these histories can be interpreted using evidence from the rock record



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